
CHAPTER

7

STATISTICAL TOOLS IN RESEARCH

Learning Outcomes

On completion of this chapter, the users will be able to

- *Know the importance of statistics in research;*
 - *Understand how to summarize collected data;*
 - *Learn about different methods of measuring central tendency, dispersion, correlation and simple and multiple regression;*
 - *Gain knowledge on the importance of normal distribution in research;*
 - *Acquire knowledge about the confidence interval.*
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7.1 What is Statistics?

Statistics is the science of collecting, analyzing and making inference from data. Statistics is particularly a useful branch of mathematics that is not only studied theoretically by advanced mathematicians but the one that is used by researchers in many fields to organize, analyze, and summarize data. Statistical methods and analyses are often used to communicate research findings and to support hypotheses and give credibility to research methodology and conclusions. It is important for researchers and also consumers of research to understand statistics so that they can be informed, evaluate the credibility and usefulness of information, and make appropriate decisions.

The word “statistics” is known to have originated from the Latin word “status” meaning “state”. For a long time, it was identified solely with the

displays of data and charts pertaining to the economic, demographic, and political situations prevailing in a country. Stretching well beyond the confines of data display, statistics now deals with collecting informative data, interpreting these data, and drawing conclusions about a phenomenon under study. The scope of this subject naturally extends to all processes of acquiring knowledge that involves finding of facts through collection and examination of data. Opinion polls, agricultural experiments, clinical studies of vaccines are just a few examples. The principles and methodology of statistics are useful in answering such questions as:

- What kind and how much data need to be collected?
- How should we organize and interpret the data?
- How can we analyze data
- How can we make use of these data to draw conclusions?
- Can we make generalization of the results so obtained?

Keeping the above questions in view, we present below a working definition of statistics:

Definition 7.1: Statistics is a subject that provides a body of principles and methodology for designing the methods of collecting, summarizing, analyzing and interpreting data, drawing valid conclusions and reaching a decision.

The use of statistics in any scientific investigation is indispensable. The detail exposition of the subject in terms of its uses and importance can be found in many texts. Our aim in this text is to make a brief overview of some statistical tools that will guide a researcher to statistically analyzing his/her data, interpreting and generalizing his/her results and then assessing the extent of uncertainty underlying these generalizations.

Two broad classifications of the subject of statistics have been made in our endeavor to present statistical methods for analyzing and interpreting the results: **descriptive statistics and inferential statistics**. We introduce the concepts of these two terms in turn.

The most common forms of descriptive statistics in use are measures of central tendency and variability of data.

Definition 7.2: Descriptive statistics are the tools which can enable us to describe large volume of data in a summarized fashion making it easy to comprehend.

When your findings are from a probability sample, summary descriptions or statistics derived from this sample may be used to infer about the corresponding population parameters under certain assumptions about the

distribution of the underlying population. This falls under inferential statistics.

Definition 7.3: Statistical procedures that allow you to infer from what you found in a representative sample to the whole population are called inferential statistics.

Such statistics may be used to test hypotheses about the relationships that may exist within a population under study. Put differently, this is done by asking whether the patterns actually found in the sample data would differ from those in the population from which the sample data were drawn.

7.2 Summarizing Data

A set of data even if modest in size, is often difficult to comprehend and interpret directly in the form in which it is collected. Suppose a survey is conducted among 154 sex workers in a community to know their status in several respects. Among others, age was considered as one of the important background variables. Table 7.1 below displays these age data in years arranged in ascending order.

If someone wants to know, for example, how many of the values are below 25 or how many are between 17 and 19, it will simply be a cumbersome job to look for this information. The problem becomes more acute and sometimes impossible if data set is bigger. What would then one expect us to do with this large volume of data? Most of us would wish that someone had classified, categorized or summarized the data in a more convenient and readily interpretable form. Such a form can be displayed by what is known as frequency distribution. We illustrate below the formation of such a distribution with the data collected in a survey among 154 sex workers in Akaka city

Table 7.1 Age Data of 154 Sex Workers Arranged in Ascending Array

12	12	13	13	13	14	15	15	15	15	15	16	16	16	16
17	17	17	17	18	18	18	18	18	18	19	19	19	19	20
20	20	20	20	20	20	20	20	20	20	20	20	20	21	21
21	21	22	22	22	22	22	22	22	22	22	22	22	22	22
22	23	23	23	23	23	23	23	23	23	23	23	23	24	24
24	24	24	24	24	24	25	25	25	25	25	25	25	25	25
25	25	25	25	26	26	26	26	26	26	26	27	28	28	28
28	28	28	28	28	28	28	29	29	29	29	30	30	30	30
30	30	30	30	30	32	32	32	33	33	34	34	34	34	35
36	37	37	38	38	38	40	40	40	42	42	42	45	45	49
50	50	65												

7.2.1 Frequency Distribution and its Construction

Among the several techniques of condensing data, frequency distribution is one of them. A frequency distribution is a simple device for arraying data. It arrays data from the lowest value to the highest, with columns for absolute and percent frequencies. Such a frequency distribution is formed in the accompanying table with the data presented in Table 7.1 and is displayed in Table 7.2.

Table 7.2: Ungrouped Frequency Distribution for Data in Table 7.1

Age	Frequency	%	Age	Frequency	%
12	2	1.29	28	10	6.49
13	3	1.95	29	4	2.60
14	1	0.65	30	9	5.84
15	5	3.25	32	3	1.95
16	4	2.60	33	2	1.29
17	4	2.60	34	4	2.60
18	6	3.90	35	1	0.65
19	4	2.60	36	1	0.65
20	14	9.09	37	2	1.29
21	4	2.60	38	3	1.95
22	14	9.09	40	3	1.95
23	12	7.79	42	3	1.95
24	8	5.19	45	2	1.29
25	13	8.44	49	1	0.65
26	7	4.54	50	3	1.95
27	1	0.65	65	1	0.65
Total	—	—	—	154	100

As you observe from Table 7.2, the distribution is still difficult to grasp since it consists of a large number of categories. An alternative way of presenting the distribution is to make a set of mutually exclusive groups of the individual ages and make a **grouped frequency distribution**. The resulting distribution is presented in Table 7.3.

Table 7.3 tells you that, of the 154 respondents, 29 (18.8%) are below age 20. Also you can tell that there are only 4 (2.6%) respondents who are over 50 years of age. In a variety of ways you can interpret the tabular data. A column for cumulative frequencies is also added in the table for future use. For example, if you desire to know how many women are under 40 years of age, you can at once say that there are 141 women who are below 40 years. This is evident from the third entry (141) in the cumulative frequency column corresponding to the age group 30–40. How many of the

women are aged 40 and above? This can be read off from the more than cumulative (also called decumulative frequency) frequency column down in the last column of the table. This is 13 which is evident from the entry against the age group 30–40.

Table 7.3: Grouped Distribution for Data in Table 7.1

Age group	Frequency (f)	Percent (%)	Cumulative frequency (CF)	Decumulative frequency (DCF)
(1)	(2)	(3)	(4)	(5)
10–20	29	18.8	29	154
20–30	87	56.5	116	125
30–40	25	16.3	141	38
40–50	9	5.8	150	13
50–60	3	2.0	153	4
60–70	1	0.6	154	1
Total	154	100	–	–

You can present the tabular data in graphic form too. Histogram is a common diagram to display data of this type. A frequency curve also can be drawn. We display the data below by a histogram (see Figure 7.1). Note that a histogram is a suitable diagram when the data are continuous in nature in which the areas of the rectangles in the histogram represent the frequencies.

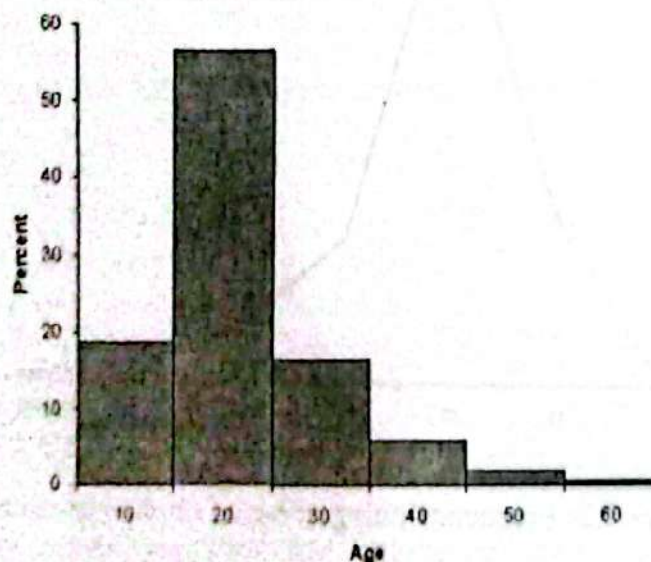


Figure 7.1: Histogram showing the age distribution in table 7.3

The histogram clearly depicts the respondents belonging to 20–30 age group constituting the largest share with about 57% of the total respondents. This is followed by their immediate younger counterpart in 10–20 age group with about 19% of the total respondents.

A popular way of representing continuous frequency distribution is to use frequency polygon.

7.2.2 Frequency Polygon

A frequency polygon provides an alternative to a histogram as a way of graphically presenting a distribution of a continuous variable. The presentation involves placing the mid-values on the horizontal axis and the frequencies on the vertical axis. However, instead of using rectangles, as with the histogram, we find the class mid-points on the horizontal axis and then plot points directly above the class mid-points at a height corresponding to the frequency of the class. Classes of zero frequency are added at each end of the frequency distribution so that the frequency polygon touches the horizontal axis at both ends of the graph. This makes the frequency polygon a closed figure. The frequency polygon is then formed by connecting the points with straight lines. We illustrate the case with the frequency distribution presented in Table 7.3 constructed from age data of 154 sex workers. The resulting ogive appears in Figure 7.2 below:

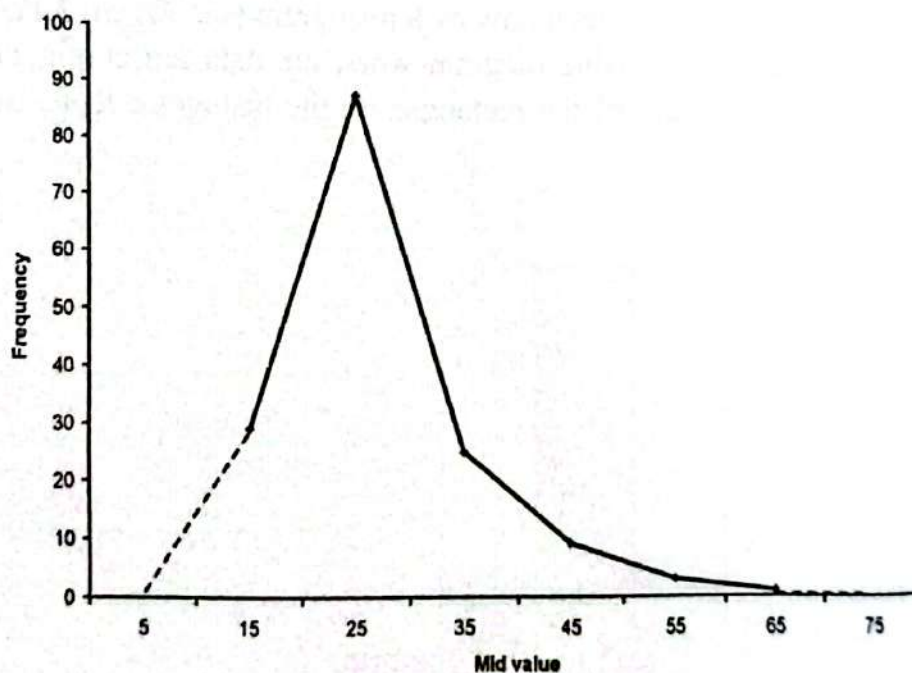


Figure 7.2: Frequency polygon based on data in table 7.3

The histogram and frequency polygon are equally good techniques for presenting continuous data. A histogram is more often used when single distributions are presented, while the frequency polygon is largely used for comparison of two or more distributions.

In a continuous frequency distribution, if the number of observations is large, then the number of classes can be increased so as to make the magnitude of class intervals smaller and smaller. And in such a case, the graph representing the distribution will approach a smooth curve. The same is true in the case of a frequency polygon too. Such a curve is called a **frequency curve**. That is, when a frequency polygon is smoothed; the resulting curve will be a frequency curve. A graph of the cumulative frequency distribution is called an **ogive**. Two forms of ogive can be constructed. One is based on the less than type cumulative frequencies and the other on the more than type cumulative frequency distribution. The fourth and the fifth columns of Table 7.3 show these cumulative frequencies, one based on the less than type cumulative frequencies and the other on more than (decumulative) frequencies. One can display these distributions by graphs also. These graphs are shown in Figure 7.3 and Figure 7.4 below:

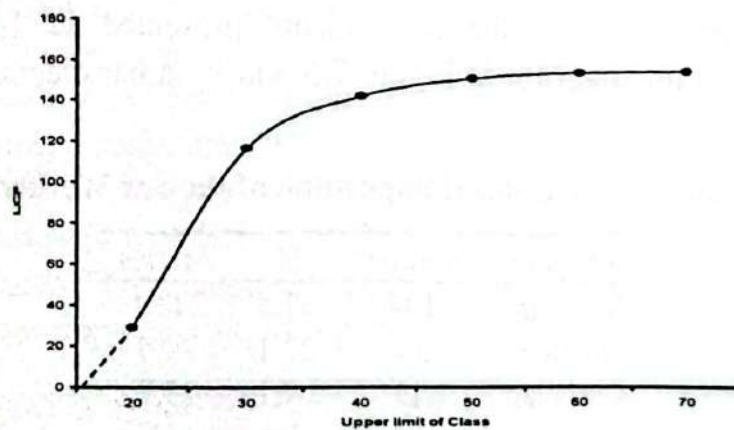


Figure 7.3: Less than type cumulative frequency polygon

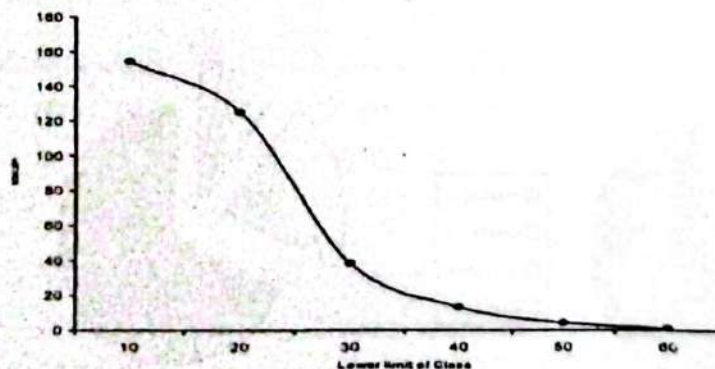


Figure 7.4: More than type cumulative frequency polygon

The histogram and frequency polygon are equally good techniques for presenting continuous data. A histogram is more often used when single distributions are presented, while the frequency polygon is largely used for comparison of two or more distributions.

To represent nominal and ordinal level data, bar and pie diagrams are almost always used. Religion (Muslim, Hindu, Christian, Buddhist, etc), region of residence (Dhaka, Chittagong, etc), for example, are nominal level data, which we can represent by bar or by a pie diagram. The following data on religious composition were derived from a survey. We represent them by bar and pie diagrams.

To represent the data by pie diagram, we need to convert the frequency in each category into angles in degrees. Thus to convert 104 into angle, for example, we proceed as follows:

$$\frac{104}{154} \times 360 = 243.1$$

As we can see, Muslims contribute the largest share to the religious composition of the respondents ((67.5%) followed by Hindus (22.1%). The religious composition of the respondents presented in Table 7.4 is displayed by a **pie diagram** in Figure 7.5 and by a bar diagram in Figure 7.6.

Table 7.4: Religious Composition of the Sex Workers

Religion	Number	%	Angles
Muslim	104	67.5	243.1
Hindu	34	22.1	79.5
Christian	11	7.1	25.7
Buddhist	5	3.3	11.7
Total	154	100	360

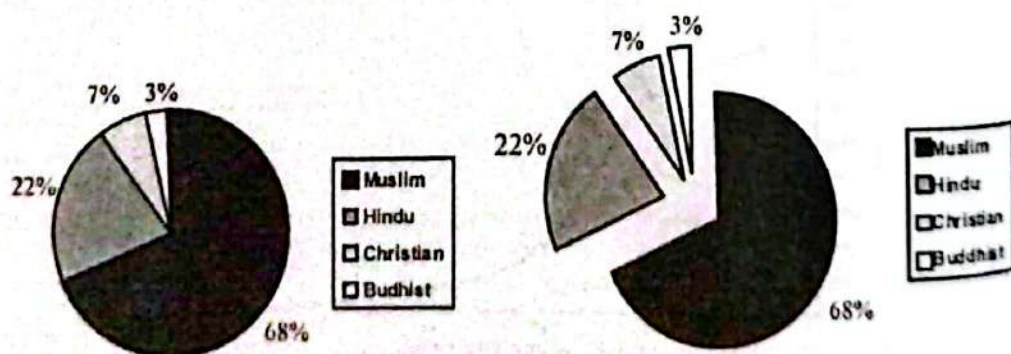


Figure 7.5: Pie diagram showing the data in table 7.4

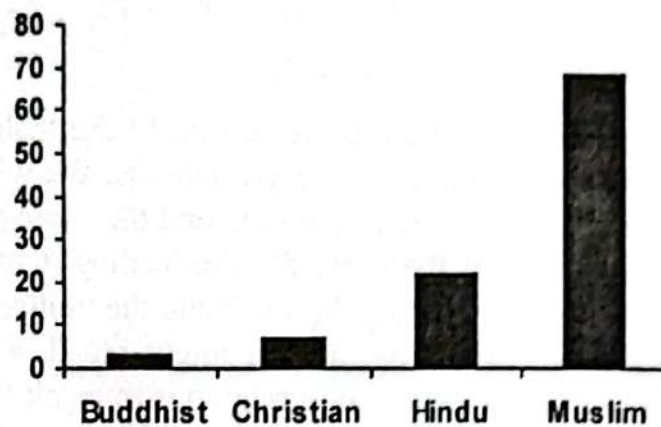


Figure 7.6: Bar chart displaying the data in table 7.4

7.2 Stem-and-leaf Plot

Stem-and-leaf plot is a graphical technique of representing quantitative data that can be used to examine the shape of a frequency distribution, the range of the values and point of concentration of the values. This is, in essence a display technique taken from the area of statistics called **exploratory data analysis (EDA)**. Compared to other graphical techniques presented thus far, stem and leaf plot is an easy and quick way of displaying data. The technique was first proposed by Tukey (1977) as an aid to understanding and exploring data through statistical analysis.

In contrast to histogram, which loses information by grouping data values into intervals, the stem-and-leaf presents actual data values that can be inspected directly, without the use of enclosed bars or asterisks as the representation medium.

The above feature reveals the distribution of values within the interval and preserves their rank order for finding the median, quartiles, and other summary measures. It also eases linking a specific observation back to the data file and to the subject that produced it.

Visualization is the second advantage of stem-and-leaf displays. It allows us to use the information contained in a frequency distribution, to show the range of score, concentrations of the scores, the shape of the distribution, presence of any specific values or scores not represented and whether there are any stray or extreme values (outliers) in the distribution.

We now illustrate the technique by an example.

Example 7.1: The following data represent the marks obtained by 20 students in a statistics test.

84	17	38	45	47	53	76	54	75	22
66	65	55	54	51	33	39	19	54	72

Use a stem and leaf plot to display the data.

Solution: We note that the lowest score is 17 and the highest is 84. For stem and leaf plots, classes must be of equal lengths. We will use the *first* or the *leading* digit (tens) of score as the stem and the *trailing* (units) digit as the leaf. For example, for the score 84, the leading digit is 8, and the trailing digit is 4; for 72, the leading digit is 7 and the trailing digit is 2 and so on. In a frequency distribution, as you might recall, a class interval determines where a measurement or observation is to be placed.

The stem and leaf plot follows the same principle, in which a leading digit (stem of a score) determines the row in which the score is placed. The trailing digits for a score are then written in the appropriate row. In this way each score is recorded in the stem and leaf plot.

With the given data now, let us take the “stem” to represent the tens (leading digits) and the “leaf” the units (trailing digits). Thus for the first 5 scores 84, 17, 38, 45, and 47, the plot is shown in the accompanying display. Next to it, following the same procedure, the complete presentation of the observations is displayed. A three-step procedure completes the presentation.

Step 1: Display for the first five observations

Stem	Leaf
1	7
2	
3	8
4	5 7
5	
6	
7	
8	4

Step 2: The complete diagram with all the observations is

Stem	Leaf
1	7 9
2	2
3	8 3 9
4	5 7
5	3 4 5 4 1 4
6	6 5
7	6 5 2
8	4

Step 3: We then arrange the leaves in ascending order in order to make the plot a bit neater and give an explanatory message or a key. The final figure is

Stem	Leaf
1	7 9
2	2
3	3 8 9
4	5 7
5	1 3 4 4 4 5
6	5 6
7	2 5 6
8	4

Key: 1|7 represents 17.

Note that each stem defines a class interval and limits of each interval are the largest and the smallest possible scores for the class. The values represented by each leaf must be between the lower and the upper limits of the interval. The chosen classes in this particular instance are seen to be 10–19, 20–29, and 80–89.

To read the score from the above figure, start at the first row and read the scores 17 and 19. These scores are shown as 1|7, 9. The key beneath the table helps to understand this presentation. The second row contains 22, while the third row contains three scores: 33, 38 and 39 and so on. Note that the number of leaves must be equal to the number of observations. From the figure, the largest score (84) and the smallest score (17) can be readily located. In addition, an entire picture of how the scores are distributed (or scattered) emerges. For example, it is readily apparent that there are more scores in the fifties than any other group; 8 scores are less than 50, and only 4 scores are above 70. Additionally, some of the numbers on the stem may have no corresponding leaves. That is in the figure, the stem position “2” would have no corresponding leave if the observation “2” were removed from the data set. For a detailed exposition of the stem and leaf plot, see Islam (2008).

Note that the plot looks like a horizontal histogram. It turns out to be a usual histogram if the plot is rotated 90 degrees counterclockwise. The advantage of a stem and leaf plot over the histogram is that it reflects not only frequencies, concentrations of scores and shape of the distribution, but also the actual score from which we can determine whether there are any values not represented and whether there are stray or extreme values (outliers). Another advantage of a stem and leaf plot is that it retains the original data.

7.3 Measures of Central Tendency

The most important aspect of studying the distribution of a sample of measurements is locating the position of a central value about which the measurements are distributed. The three most commonly used indicators of centre of a distribution are mean, median and mode. These indicators are numerical values that attempt to answer the question: what is the typical value of the measurements in this distribution? They are generally indicative of the 'centre' of the set. It is because of this reason such measures are called 'measures of central tendency'.

7.3.1 Arithmetic Mean

The arithmetic mean of a set of measurement is the sum of the measurements divided by the number of such measurements. It is the location measure most frequently used for interval and ratio level data. For instance, the mean for five measurements 2, 3, 6, 7, 12 is

$$\frac{2+3+6+7+12}{5} = \frac{30}{5} = 6$$

To state this concept in general terms, we use symbols. If a sample consists of n measurements symbolized $x_1, x_2, \dots, x_n$, and we use $\bar{x}$ (pronounced x bar) to represent the sample mean of these n measurements, then

$$\bar{x} = \frac{x_1 + x_2 + \dots + x_n}{n} \quad \dots(1)$$

Using a Greek capital letter sigma Σ , we write the above expression as follows:

$$\bar{x} = \frac{\sum x}{n} \quad \dots(2)$$

The mean can be most misleading when the distribution contains extreme scores, large or small. Such extreme scores are called **outliers**.

For a frequency distribution, the computational process of the mean is a bit different. We illustrate the computational procedure by an example.

Example 7.2: Suppose we have a distribution of 100 customers by the amount ('000) they withdrew from a local bank on a specified day:

Amount (x)	Number of customers (f)	Product (fx)
10	5	50
12	20	240
15	40	600
16	25	400
17	10	170
Total	100	1460

What is the average amount the customers withdrew from the bank?

The arithmetic mean or average for such frequency distribution is calculated in a 3-step procedure:

1. Multiply each value (x) in column 1 by its corresponding frequency (f) in column (2). This is fx , which is shown in column (3)
2. Sum the product obtained in step 1. This is $\sum fx (=1460)$.
3. Divide the resulting sum obtained in step 2 by the total frequency ($\sum f=100$). This gives you the mean of the distribution ($\bar{x}$).

The general formula for computing mean of a frequency distribution is thus

$$\bar{x} = \frac{\sum fx}{\sum f} = \frac{\sum fx}{N} \quad \dots(3)$$

Using the formula, the mean amount is

$$\bar{x} = \frac{(5 \times 10) + (20 \times 12) + (40 \times 15) + (25 \times 16) + (10 \times 17)}{100} = \frac{1460}{100} = 14.6 \text{ thousand}$$

When the volume of data is large, it is rewarding to put them in a condensed form, which we call grouped data. The raw data are put in an arrayed form to facilitate grouping. When such grouping is done, the resulting table is a grouped frequency distribution.

Example 7.3: Compute the arithmetic mean for the data presented in Table 7.1. The usual procedure to compute mean of this data set is to add all the values and dividing the resulting sum by the total number of observations (here 154). That is

$$\bar{x} = \frac{x_1 + x_2 + \dots + x_n}{n} = \frac{12 + 12 + \dots + 65}{154} = \frac{3965}{154} = 25.75$$

The process is no doubt cumbersome. A somewhat convenient solution to this problem is to make a frequency distribution of the form as shown in Table 7.2 without grouping. We reproduce the table here for computational convenience and carry out the computation following Example 7.2.

Age	Frequency	Age	Frequency
12	2	28	10
13	3	29	4
14	1	30	9
15	5	32	3
16	4	33	2
17	4	34	4
18	6	35	1
19	4	36	1
20	14	37	2
21	4	38	3
22	14	40	3
23	12	42	3
24	8	45	2
25	13	49	1
26	7	50	3
27	1	65	1

The mean now can be recalculated as

$$\bar{x} = \frac{(12 \times 2) + (13 \times 3) + \dots + (50 \times 3) + (65 \times 1)}{2 + 3 + \dots + 3 + 1} = \frac{3965}{154} = 25.75$$

as before.

The computation is still tedious owing to the presence of too many groups. A neat solution to the problem is to summarize the data by forming classes resulting in a grouped frequency distribution. One such possible distribution has been constructed earlier and presented in Table 7.3. We reproduce the table here:

Age group	Mid-values (x)	Frequency (f)	Product (fx)
10-20	15	29	435
20-30	25	87	2175
30-40	35	25	875
40-50	45	9	405
50-60	55	3	165
60-70	65	1	65
Total	—	154	4120

The first step in computing mean from this table is to locate the mid-points of classes and then multiply these mid-points by the frequencies and then sum them. The resulting sum is then divided by the total frequency to obtain the mean. Thus, using (3)

$$\bar{x} = \frac{\sum fx}{\sum f} = \frac{4120}{154} = 26.75$$

Note that the mean computed from the group distribution is different from that computed from the raw data (Table 7.1). It is therefore always advisable to compute the mean from the raw data whenever possible where any transformation is made in the data.

Though this form of presentation has greatly condensed the data, the individual identity of the observations has been lost. In addition, the mean and other descriptive measures computed from this distribution will only be an approximation.

If analysis of data demands comparison of the means by sub-categories of the respondents (say by religion, socio-economic classes etc.), one can compute means for each such sub-group separately. Suppose we have two groups of observations. Suppose further that the first group has n_1 observations with a mean $\bar{x}_1$. The second group has n_2 observations with a mean $\bar{x}_2$. The mean of these two groups of observations together, called combined or pooled mean is $\bar{x}_c$, which we compute as follows:

$$\bar{x}_c = \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2}{n_1 + n_2} \quad \dots (4)$$

Example 7.4: Suppose we have two sets of observations as follows:

Set I: 5, 10, 15, 7, 13 Set II: 10, 4, 5, 9, 8, 7, 6, 10, 5, 6

If we compute the mean of the two sets of 15 observations together, the mean is

$$\bar{x}_c = \frac{(5+10+\dots+13)+(10+4+\dots+6)}{15} = \frac{120}{15} = 8$$

The means of the individual sets are

$$\bar{x}_1 = \frac{5+10+\dots+13}{5} = \frac{50}{5} = 10$$

$$\bar{x}_2 = \frac{10+4+\dots+6}{10} = \frac{70}{10} = 7$$

The combined or pooled mean using (4) is

$$\bar{x}_c = \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2}{n_1 + n_2} = \frac{(5 \times 10) + (10 \times 7)}{5 + 10} = \frac{120}{15} = 8$$

as aught to be. Note that simple mean of the two means does not lead to the correct mean: $(10+7)/2=8.5$, while the correct mean is 8.

Our data in Table 7.1, for example, have two categories of respondents: brothel-based sex workers (BSW) and street sex workers (SSW). Using a computer program, we computed the group means, which appear below together with the number of cases:

Respondent type	Number	Mean age
BSW	71	28.13
SSW	83	23.71
Total	154	25.75

You can easily demonstrate that the overall mean can be computed from the group means knowing the number of cases in each group.

$$\bar{x}_c = \frac{(71 \times 28.13) + (83 \times 23.71)}{154} = 25.75$$

which exactly agrees with the one computed from the ungrouped data.

This can be done for any number of sub-categories. The general formula for k sub-categories is

$$\bar{x}_c = \frac{n_1 \times \bar{x}_1 + n_2 \times \bar{x}_2 + \dots + n_k \times \bar{x}_k}{n} = \frac{\sum n_i \bar{x}_i}{\sum n_i} \quad (i=1, 2, \dots, k) \quad (5)$$

A similar concept may be employed to compute the mean of a different type, called **weighted mean**. This mean is used to average rates, ratios, proportions and percentages. We illustrate the use of this mean by an example.

Example 7.5: Suppose you make a purchase of 5 items at the rate of taka 20 and another 7 items at the rate of taka 25. What is your average rate for these 12 items?

Solution: This is a problem of weighted mean, which we denote by $\bar{x}_w$ and compute it as

$$\bar{x}_w = \frac{r_1 w_1 + r_2 w_2}{w_1 + w_2} = \frac{(20 \times 5) + (25 \times 7)}{5 + 7} = 22.92$$

Example 7.6: The prevalence of malnutrition among 780 under-5 male children is 0.05 while this rate is 0.03 among 1000 female children. What is the overall rate for all the children under study?

Solution: Here $r_1=0.05$, $r_2=0.03$, $w_1=780$, and $w_2=1000$, so that

$$\bar{x}_w = \frac{r_1 w_1 + r_2 w_2}{w_1 + w_2} = \frac{(.05 \times 780) + (.03 \times 1000)}{780 + 1000} = 0.0388$$

Following (5), you can extend the formula to k sub-groups:

$$\bar{x}_w = \frac{\sum r_i w_i}{\sum w_i} \quad (i=1, 2, \dots, k) \quad \dots(6)$$

7.3.2 Geometric Mean

Geometric mean is a rarely used measure of centrality of data. It is used with numbers that tend to increase geometrically rather than arithmetically, that is, each number is the same multiple of the preceding number. It is typically used in averaging index numbers, rates of change, ratios, and other sets of values expressed in ratio or percentage forms.

Geometric mean has limited applications in business and economics, where we are often interested in determining the percentage changes in sales, GNP and the like. For n non-zero positive quantities $x_1, x_2, \dots, x_n$, the geometric mean G is defined as the product of all the values of x :

$$G = \sqrt[n]{x_1 \times x_2 \times \dots \times x_n} \quad \dots(7)$$

That is, to calculate geometric mean, we multiply all the n values and extract n^{th} root of the product. Thus for two values 2 and 8, for example

$$G = \sqrt{2 \times 8} = \sqrt{16} = 4$$

Similarly, for $n=3$ values, G is the cube root of the product of x_1, x_2 and x_3 :

$$G = \sqrt[3]{x_1 \times x_2 \times x_3}$$

For a frequency distribution, the geometric mean for k classes is defined by the equation

$$G = \sqrt[n]{x_1^{f_1} \times x_2^{f_2} \times \dots \times x_k^{f_k}}$$

where $n = \sum f_i$

Example 7.7: An individual had a monthly salary of Tk. 2,000 in 1980, which increased to Tk. 4,000 in 1990, with a further increase to Tk. 18,000 in the year 2000. Find an appropriate average for the 10-year period.

Solution: Note that the salary has increased 2 times from its initial amount of Tk. 2,000 in 1980 to Tk. 4,000 in 1990 and to Tk. 18,000 in the year 2000. Pretty obvious, the salary has not been increasing arithmetically. We check this as follows:

The rates of change have been 2 and 4.5, the arithmetic mean of which is 3.25. With this increase, the estimated salary for 1990 is $2000 \times 3.25 = 6,500$ and for the year 2,000, it is $6,500 \times 3.25 = 21,125$. Neither of these estimated salaries compares favorably with the actual values of 4,000 and 18,000. We do not therefore recommend arithmetic mean to be used here as an appropriate average. The geometric mean of 2 and 4.5 is $\sqrt{2 \times 4.5} = 3$, which, when applied to the given data, yields an estimated salary of $2000 \times 3 = 6,000$ for 1990 and $6,000 \times 3 = 18,000$ for 2000. These estimates compare well with the given values. Thus we employ geometric mean in the present instance:

$$G \doteq \sqrt[3]{2,000 \times 4,000 \times 18,000} = Tk.5241$$

In computing G, the use of logarithm appears to be highly rewarding in many instances. We illustrate how the computation can be accomplished with the salary data. Designating the 3 years' salary values by x_1, x_2 and x_3

$$G = \sqrt[3]{x_1 x_2 x_3}$$

Taking logarithm,

$$\begin{aligned} \text{Log } G &= \frac{\log x_1 + \log x_2 + \log x_3}{3} \\ &= \frac{\log_{10}(2,000) + \log_{10}(4,000) + \log_{10}(18,000)}{3} \\ &= \frac{3.30103 + 3.60206 + 4.25527}{3} \\ &= 3.71945 \end{aligned}$$

Taking antilog of 3.71945

$$G = 5241 \text{ as before.}$$

The above example demonstrates why the mean of the logarithms of a set of numbers is significantly less influenced by extreme values than the mean of the original arithmetic values. Occasionally, an analyst prefers to use the geometric mean in order to achieve precisely this effect.

Example 7.8: Suppose a piece of properties is purchased at Tk. 2,00,000 and sold 10 years later at Tk. 3,20,000. What is the average annual return on the original Tk. 2,00,000 investment?

Solution: Let G denote the geometric mean. Then

$$2,00,000G^{10} = 3,20,000$$

$$G^{10} = 1.6$$

Thus

$$\log G = \frac{\log 1.6}{10} = \frac{0.2041}{10} = 0.02041$$

Taking antilog

$$G = 1.048$$

Thus the investment yields a mean rate of return of 4.8 percent over the 10-year period. Check that the arithmetic mean rate of return is 6 percent.

The geometric mean is unique and employs all the observations. It assigns less weight to extreme values. However, it cannot be used when the data set contains a 0 or one or more negative values. It is difficult to compute and at the same time difficult to conceptualize also. For all non-zero positive values, it distinctively remains less than the arithmetic mean but greater than harmonic mean.

7.3.3 Harmonic Mean

Harmonic mean (H) is also a rarely used measure of central tendency. For a set of n non-zero positive values $x_1, x_2, \dots, x_n$, the harmonic mean is the reciprocal of the arithmetic mean of the reciprocal of the values. It is defined by the equation

$$H = \frac{n}{(1/x_1) + (1/x_2) + \dots + (1/x_n)} = \frac{n}{\sum (1/x_i)} \quad \dots(8)$$

In practice, we compute

$$\frac{1}{H} = \frac{1}{n} \left(\frac{1}{x_1} + \frac{1}{x_2} + \dots + \frac{1}{x_n} \right) = \frac{1}{n} \sum \left(\frac{1}{x_i} \right) \quad \dots(9)$$

and invert it to obtain H . For example, if $1/H$ is 0.25, then H is 4.

The application of harmonic mean is illustrated by an example below.

Example 7.9: Suppose that a person spent Tk. 100 in each of the three fruit shops. In the first shop, he bought orange at 4 taka a piece; in the second shop, he bought orange at 5 taka a piece; and in the third shop, he bought orange at 10 taka each. What is the average price he paid per piece of orange?

Solution: The data are expressed as 'so many pieces of orange in 100 taka' while we wish to know the amount of money paid per piece of orange. We compute the arithmetic mean

$$A = \frac{4 + 5 + 10}{3} = 6.33$$

we will be in error because more oranges were bought at 4 taka than at 5 taka and more were bought at 5 taka than at 10 taka. In other words, a weighted average is required. Since the person could buy 25 oranges from the first shop, 20 from the second shop and 10 from the third shop, the weighted average is

$$WA = \frac{25 \times 4 + 20 \times 5 + 10 \times 10}{25 + 20 + 10} = 5.45$$

We check that the harmonic mean provides the same value:

$$\frac{1}{H} = \frac{1}{3} \left(\frac{1}{4} + \frac{1}{5} + \frac{1}{10} \right) = \frac{11}{60}$$

Inverting, the harmonic mean H is

$$H = \frac{60}{11} = 5.45$$

Example 7.10: An investor purchases 10 shares from Company A worth taka 1000 and another 16 shares from Company B with the same amount. How much did he pay on average per share?

Solution: The investor paid a total of 2000 taka in purchasing 26 shares. Therefore, the average is simply $2000/26 = \text{Tk. } 76.92$ per share. The same average can be obtained by using harmonic mean. For this we compute the prices per share for the two companies:

Company	Amount paid	No. of shares purchased	Price per share
A	1,000	10	100
B	1,000	15	62.5
Total	2,000	25	—

$$\frac{1}{H} = \frac{1}{2} \left(\frac{1}{100} + \frac{1}{62.5} \right) = 0.013$$

This gives

$$H = 76.92.$$

Solution: The data are expressed as 'so many pieces of orange in 100 taka while we wish to know the amount of money paid per piece of orange' we compute the arithmetic mean

$$A = \frac{4+5+10}{3} = 6.33$$

we will be in error because more oranges were bought at 4 taka than at 5 taka and more were bought at 5 taka than at 10 taka. In other words, a weighted average is required. Since the person could buy 25 oranges from the first shop, 20 from the second shop and 10 from the third shop, the weighted average is

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We check that the harmonic mean provides the same value:

$$\frac{1}{H} = \frac{1}{3} \left(\frac{1}{4} + \frac{1}{5} + \frac{1}{10} \right) = \frac{11}{60}$$

Inverting, the harmonic mean H is

$$H = \frac{60}{11} = 5.45$$

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This gives

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7.3.4 Median

Median is the mid-point of the distribution. Half of the observations in the distribution fall above and the other half fall below the median. For example, the data set 6, 9, 12, 15, 18 has a median value 12, because 12 stands on the middle when arranged from smallest to largest or from largest to smallest. When the distribution has an even number of observations, the median is the average of the two middle scores. The set 3, 5, 7, 8 has two middle values 5 and 7, so that median = $(5+7)/2=6$. For the sex workers data, the median is the average of the 57th and 58th values, since the data set consists of an even number of cases. You can check that both these are 22. Hence the median is 22. We will use $\bar{M}$ to denote the median.

The median is the value that divides an ordered set of n observations into two equal parts. For an odd number of values, it is the central value:

$$\bar{M} = \left(\frac{n+1}{2} \right) \text{th value} \quad \dots(10)$$

For an even number of values, the median is the arithmetic mean of the two central values:

$$\bar{M} = \text{Average of } \left[\frac{n}{2} \text{ and } \left(\frac{n}{2} + 1 \right) \right] \text{th values} \quad \dots(11)$$

The definition implies that the observations must be arranged in order of their magnitude before locating median. The same principle is applicable for grouped data but is based on an algebraic equation of the form

$$\bar{M} = l_o + \frac{h}{f_m} \left(\frac{n}{2} - F_{-m} \right) \quad \dots(12)$$

where

- l_o = Lower limit of the median class
- h = Width of the median class
- f_m = Frequency of the median class
- F_{-m} = Cumulative frequency of the pre-median class
- n = Total frequency

We illustrate the use of this formula using the data in Table 7.3.

Age Group	Frequency (<i>f</i>)	Cumulative frequency
10–20	29	29 (<i>F_m</i>)
20–30	87 (<i>f_m</i>)	116
30–40	25	141
40–50	9	150
50–60	3	153
60–70	1	154
Total	154	–

In this example, $n=154$ so that $n/2=77$. Looking down the cumulative column, we find that 77th value falls in the age range 20–30 so that the median class is 20–30. Further more $l_0=20$ and $h=10$. Hence

$$\bar{M} = l_0 + \frac{h}{f_m} \left(\frac{n}{2} - F_{-m} \right) = 20 + \frac{10}{87} \left(\frac{154}{2} - 29 \right) = 25.5$$

Mean or the median: which one to choose?

Mean is the most commonly used measure of central value. It is the most amenable to analysis but can be misleading when data are not distributed symmetrically about a central value. The median is the most appropriate locator of centre for ordinal data and has resistance for extreme values (outliers), thereby making it a preferred measure for interval and ratio level of data—particularly those with asymmetric distributions. How to choose between median and mean? Consider the following ordered data set which represents the number of days the six heart transplant patients at a hospital survived after their operations:

10, 25, 46, 64, 126, 623

There are two middle values, 46 and 64, so that

$$\bar{M} = \frac{46 + 64}{2} = \frac{110}{2} = 55 \text{ days}$$

The mean is

$$\bar{x} = \frac{10 + 25 + 46 + 64 + 126 + 623}{6} = \frac{894}{6} = 149 \text{ days}$$

Note that one survival time greatly inflates the mean. Only 1 out of 6 patients survived longer than the mean value. Here the median of 55 days appears to be better indicator of the center than the mean.

The example demonstrates that the median is not affected by outliers, whereas the presence of such values can have a considerable effect on the mean. That is why government reports on income distribution quote the

median income as a summary, rather than the mean. A relatively small number of very highly paid persons can have a great effect on the mean salary.

7.3.5 Trimmed Mean

The effect of extreme values can also be avoided by applying what is known as **trimmed mean**. A trimmed mean is computed by trimming away a certain portion of the data set when the set contains unusually extreme values. In the above data set, we have an extreme value 623 which unusually inflates the mean. Discarding this value from the set, we obtain a trimmed mean (TM) which is closed to the median.

$$TM = \frac{10 + 25 + 46 + 64 + 126}{5} = \frac{266}{5} = 53.2$$

7.3.6 Quartiles

Other measures that are allied to the median include the **quartiles** and **percentiles**, because they are also based on their position in a series of observations. They together are variously called **quantiles**, **fractiles** or **partition values**. These measures are not necessarily central values and hence they are referred to as **measures of location** and not the measures of central tendency. We discuss these measures below.

There are three quartiles in a data series, usually denoted by Q_1 , Q_2 and Q_3 , which divide the whole distribution into four equal parts. The second quartile Q_2 , is identical with the median. The first quartile, Q_1 , is the value at or below which one-fourth (25%) of all items in the series fall; the third quartile, Q_3 is the value at or below which three-fourths (75%) of the item lie.

With ungrouped data, a quartile, as does the median, either assumes the value of one of the items or falls between two values. If n is divisible by 4, the first quartile (Q_1) has the value half-way between the $n/4$ -th and $(n/4 + 1)$ -th number. If n is not exactly divisible by 4 (i.e. $n/4$ is not an integer), the first quartile has the value of the next higher integer. To find the third quartile Q_3 , we replace $n/4$ by $3n/4$.

Consider the following set of 12 values arranged in ascending order:

14, 17, 19, 23, 27, 32, 40, 49, 54, 59, 71 80.

Here $n = 12$, which is divisible by 4. The quotient is 3. Thus the first quartile will be the average value of the 3rd and the 4th items:

$$Q_1 = \frac{19 + 23}{2} = 21$$

We add a new value 94 to the set so that $n/4$ is not an integer. Here $n/4 = 13/4 = 3.25$. The next higher integer is 4. Thus the 4th value will be the first quartile. Observe that the 4th value in the set is 23. Thus

$$Q_1 = 4\text{th value} = 23$$

With $n=12$, the third quartile is the value mid-way between $3n/4$ -th and $(3n/4+1)$ th observations, since $3n/4 = 9$ is an integer. Thus Q_3 is the average of the 9th and 10th observations:

$$Q_3 = \frac{54 + 59}{2} = 56.5$$

Adding 94 in the series n becomes 13 and the third quartile now is the 10th observation, since $3n/4 = 39/4 = 9.75$, which is not an integer. The next higher integer is 10. Thus Q_3 is the 10th value, which is equal to 59.

For large data set, the computation of quartile values simply by inspection of the data set is practically impossible. In such situations the quartile values can be calculated by first forming the cumulative frequency distribution and then locating the desired quartile from the given values based on n , the number of observations. The following is an example, which illustrates how to compute median from such data.

Example 7.11: Distribution of 70 students according to the marks they obtained in a class test is as follows:

Marks	No. of students	Cumulative frequencies
40	6	6
43	11	17
51	19	36
55	17	53
60	13	66
63	4	70
Total	70	-

To obtain Q_1 and Q_3 , we cumulate the frequencies as shown above. Since $n/4 = 70/4 = 17.5$ is not an integer, the first quartile will be the 18th observation (next higher integer of the fraction 17.5). From the cumulative total, Q_1 will be located, which in this case corresponds 51. Since $3n/4 = 52.5$, the Q_3 is the 53rd value which equals 55.

With grouped data, the method of estimating the first and the third quartiles is similar to that of estimating the median. This can be accomplished through the following formula:

$$Q_i = l_i + \frac{h}{f_i} \left(\frac{in}{4} - F_i \right)$$

where

l_i = Lower limit of the i -th quartile class

n = Total frequency

h = Class width of the i -th quartile class

F_i = Cumulative frequency of the class prior to the i -th quartile class

f_i = Frequency of i -th quartile class

Example 7.12: Compute Q_1 and Q_3 for the data in Table 7.3.

Solution: Following the formula above, the first quartile is

$$Q_1 = l_1 + \frac{h}{f_1} \left(\frac{n}{4} - F_1 \right)$$

To compute Q_1 , we set $i=1$ in the formula. Here $l_1=20$, $n/4=38.5$, $h=10$, $f_1=87$ and $F_1=29$.

$$Q_1 = 20 + \frac{10}{87} (38.5 - 29) = 21.1$$

To compute Q_3 , we set $i=3$ in the formula

$$Q_3 = l_3 + \frac{h}{f_3} \left(\frac{n}{4} - F_3 \right)$$

Here $l_3=20$, $3n/4=115.5$, $h=10$, $f_3=87$ and $F_3=29$.

$$Q_3 = 20 + \frac{10}{87} (115.5 - 29) = 29.9$$

A value of 21.1 for Q_1 implies that 25% of the women of the survey population are under 21.1 years. Similarly, 75 percent of all women are below 29.9 years.

7.3.7 Percentiles

The statistical measure referred to as a **percentile** offers a means for identifying the location of values in the data set that are not necessarily central values and thus may be regarded as another measure of location. This measure provides information regarding how the data items are spread over the interval from the lowest to the highest values. Hence percentiles can also be viewed as measures of dispersion or variability in the data set. In large data sets that do not have numerous repeat values, the p th percentile is a value that divides the data set in two parts.

Approximately $p\%$ of the items take on values less than the p th percentile, and approximately $(100-p)\%$ of the items take on greater values.

Admission test scores for colleges and universities are frequently reported in terms of percentiles. For example, suppose an applicant has a raw score of 54 in the oral portion of an admission test. It may not be readily apparent how this student performed relative to other students taking the same test. However, if the raw score of 54 corresponds to the 70th percentile, it is easily seen that approximately 70 percent of the students had a score less than this individual and approximately 30 percent scored better.

It seems obvious that percentiles are the values, which divide the distribution into 100 equal parts. Thus there are 99 percentiles in a distribution, which are conventionally denoted by $P_1, P_2, \dots, P_{99}$.

Recall that in the discussion of the median, we found that the median divides the items arranged in order of magnitude into two equal parts. Thus in terms of percentiles the median is the 50th percentile. This means that $P_{50} = Q_2 = \text{Median}$. At times the 25th percentile and /or the 75th percentile may be of particular interest. These two percentiles are in fact the first quartile (Q_1) and third quartile (Q_3) respectively, which we discussed earlier.

With ungrouped data, the percentile either takes the value half-way between the two observations or the value of one of the observations, depending on whether n is divisible by 100 or not. Consider the observations 11, 14, 17, 23, 27, 32, 40, 49, 54, 59, 71 and 80. To determine the 29th percentile, P_{29} say, we note that $\frac{1}{100}(29 \times 12) = 3.48$, which is not an integer. Thus the next higher integer 4 and hence the 4th observation will determine the 29th percentile value. On inspection, $P_{29} = 23$. Similarly P_{75} will be the average of the 9th and 10th observations, since $(75 \times 12)/100 = 9$, which is an integer. Thus

$$P_{75} = \frac{54 + 59}{2} = 56.5$$

If the percentile values are required for a frequency distribution, the procedure adopted in computing median or quartiles may be followed. In general, the i -th percentile of a grouped distribution for n observations may be arrived at by using the following formula:

$$P_i = l_i + \frac{h}{f_i} \left(\frac{in}{100} - F_i \right)$$

where

l_i = Lower limit of the i -th percentile class

n = Total frequency

h = Class width of the i -th percentile class

F_i = Cumulative frequency of the class prior to the i -th percentile class

f_i = Frequency of the i -th percentile class

As an illustration, we compute the 85th percentile for the distribution presented in Table 7.3. The required percentile class is determined from $85n/100 = (85 \times 154)/100 = 130.9$. Looking at the cumulative frequency in the table, we find that this value falls in the range 30–40. The other required values are: $l_{30} = 30$, $h = 10$, $f_{85} = 25$ and $F_{85} = 116$. Hence

$$P_{85} = 30 + \frac{10}{25}(130.9 - 116) = 36$$

The computed percentile value implies that only 15% women are above 36 years.

7.3.8 Percentile Rank

Going back to our example of admission score of 54 just cited, the applicant might ask himself “how does my score 54 ranks me among all my fellow friends who took part in the test?”. The answer is that he scored the same as or better than 70% of the entire group, indicating that his **percentile rank** is 70%. This is to say that a score of 54 has a percentile rank of 70 indicating that 70% of the students have scored 54 or below. This example shows that knowing the percentile rank of a score or value, allows one to compare it with other scores or values.

Definition: The percentile rank of any score or observation is defined as the percentage of cases in a distribution that falls at or below that score.

Percentile ranks are simple to calculate if the entire collections of raw scores are available. For example, in the following collection of 20 scores, the applicant's 54 would be ranked 14th from the bottom:

19, 22, 25, 30, 38, 39, 41, 43, 44, 47, 48, 49, 51, 54, 56, 59, 61, 65, 67, 70.

Thus his percentile rank is **fourteenth** out of 20 or 70%.

7.3.9 Mode

The mode is the most frequently occurring value in a distribution. A distribution may be bimodal, tri-modal or even multimodal depending on whether two, three or more scores have equal frequency in the distribution.

The data set 2, 3, 4, 3, 5, 2, 3, 3, 5, 3, 3, 6, 3, has mode 3 because the score 3 occurs with the highest frequency (here 3 occurs 7 times, which is the highest). The sex workers distribution (see Table 7.2) is bimodal, because 20 and 22 occur with the same frequency ($f=14$). Mode is applicable to all levels of measurement, nominal, ordinal, interval and ratio.

Definition 7.4: Mode is the value that occurs with the greatest frequency in a set of observations.

Mode is often undefined or ambiguous for which it is used little. For grouped distribution, the mode is computed from the following formula:

$$M_O = l_o + h \left(\frac{f_0 - f_{-1}}{2f_0 - f_{-1} - f_{+1}} \right) \quad \dots(13)$$

where

- l_o = Lower limit of the modal class
- h = Width of the modal class
- f_{-1} = Frequency of the pre-modal class
- f_{+1} = Frequency of the post-modal class
- n = Total frequency

Example 7.13: Compute modal value from the frequency distribution presented in Table 7.3.

Solution: Here the modal class is 20–30 since it consists the highest frequency so that $l_o = 20$, $h = 10$, $f_{-1} = 29$ and $f_{+1} = 25$. The mode is thus

$$M_O = 20 + 10 \left(\frac{87 - 29}{174 - 29 - 25} \right) = 24.8$$

7.3.10 Empirical Mode

Provided the departure from symmetry is moderate, the relationship between the values of the mean, median, and mode are described by the following relationship:

$$M_o = \bar{x} - 3(\bar{x} - \bar{M}) \quad \dots(14)$$

Given a previous determination of the values of the mean and the median, the above relationship can be used to generate an estimate of the values of the mode. With the data in Table 7.3, the empirical formula gives 23 as the value of the mode:

$$M_o = 26.75 - 3(26.75 - 25.5) = 23$$

while the direct formula (13) yields 24.8. This departure is due to the fact that the distribution is far from symmetric.

7.4 Measures of Variation

The common measures of variations are the range, variance, inter-quartile range and quartile deviation. These measures describe how individual scores cluster or scatter in a distribution around the central value.

7.4.1 Variance

The **variance** is the average of the squared deviation scores from the arithmetic mean of the distribution. It is a measure of data or scores' dispersion about the mean. If all the scores are identical, the variance is zero. The greater the dispersion of scores, the greater is the variance. The positive square root of the variance is called the **standard deviation**. Interval and ratio level data are needed to compute variance and standard deviation. For a set of n observations $x_1, x_2, \dots, x_n$, the variance, denoted by s^2 is usually computed using the formula

$$s^2 = \frac{\sum (x - \bar{x})^2}{n}$$

When we are dealing with a sample, we use $n-1$ in the denominator instead of n , and call it sample variance:

$$s^2 = \frac{\sum (x - \bar{x})^2}{n-1}$$

For all computational purposes, the above formula can be rewritten as

$$s^2 = \frac{\sum x^2 - n\bar{x}^2}{n-1}$$

The standard deviation (s) can be computed as

$$s = \sqrt{\frac{\sum x^2 - n\bar{x}^2}{n-1}}$$

Example 7.14: Compute variance and standard deviation for the following data set 6, 9, 12, 15, 18.

Solution: The arithmetic mean for the data is

$$\bar{x} = \frac{6+9+12+15+18}{5} = 12$$

Hence the variance is

$$s^2 = \frac{(6-12)^2 + (9-12)^2 + (12-12)^2 + (15-12)^2 + (18-12)^2}{5-1} = \frac{90}{4} = 22.5$$

Using the alternative formula with $(n-1)$ as divisor

$$s^2 = \frac{(6^2 + 9^2 + \dots + 18^2) - 5(12^2)}{5-1} = \frac{90}{4} = 22.5 \text{ as before}$$

And the standard deviation is

$$s = \sqrt{22.5} = 4.74$$

For a frequency distribution with k classes, the formula for computing variance is modified as follows:

$$s^2 = \frac{\sum f(x - \bar{x})^2}{n-1} = \frac{\sum fx^2 - n\bar{x}^2}{n-1}$$

For the frequency distribution in Table 7.3 with x as mid-point, we can compute the variance as follows:

Age group	Frequency f	Mid-values x	Product fx	fx^2
10-19	29	15	435	6525
20-29	87	25	2175	54375
30-39	25	35	875	30625
40-49	9	45	405	18225
50-59	3	55	165	9075
60-69	1	65	65	4225
Total	154	—	4120	123050

$$\bar{x} = \frac{\sum fx}{n} = \frac{4120}{154} = 26.75$$

$$s^2 = \frac{\sum fx^2 - n\bar{x}^2}{n-1} = \frac{123050 - 154(26.75^2)}{153} = 84.01$$

Hence the standard deviation is

$$s = \sqrt{84.01} = 9.16$$

7.4.2 Standard Error

The standard deviation of a sample mean is often called standard error of the mean or simply standard error ($s_{\bar{x}}$). In other words, the term 'standard error' applies to mean, unless otherwise specified. The standard error is defined as

$$s_{\bar{x}} = \sqrt{\frac{\text{Sample variance}}{\text{Sample size}}} = \sqrt{\frac{s^2}{n}} = \frac{s}{\sqrt{n}}$$

For the sex worker data, the standard error is

while the direct formula (13) yields 24.8. This departure is due to the fact that the distribution is far from symmetric.

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The common measures of variations are the range, variance, inter-quartile range and quartile deviation. These measures describe how individual scores cluster or scatter in a distribution around the central value.

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The **variance** is the average of the squared deviation scores from the arithmetic mean of the distribution. It is a measure of data or scores' dispersion about the mean. If all the scores are identical, the variance is zero. The greater the dispersion of scores, the greater is the variance. The positive square root of the variance is called the **standard deviation**. Interval and ratio level data are needed to compute variance and standard deviation. For a set of n observations $x_1, x_2, \dots, x_n$, the variance, denoted by s^2 is usually computed using the formula

$$s^2 = \frac{\sum (x - \bar{x})^2}{n}$$

When we are dealing with a sample, we use $n-1$ in the denominator instead of n , and call it sample variance:

$$s^2 = \frac{\sum (x - \bar{x})^2}{n-1}$$

For all computational purposes, the above formula can be rewritten as

$$s^2 = \frac{\sum x^2 - n\bar{x}^2}{n-1}$$

The standard deviation (s) can be computed as

$$s = \sqrt{\frac{\sum x^2 - n\bar{x}^2}{n-1}}$$

Example 7.14: Compute variance and standard deviation for the following data set 6, 9, 12, 15, 18.

Solution: The arithmetic mean for the data is

$$\bar{x} = \frac{6+9+12+15+18}{5} = 12$$

Since the variance is

$$s^2 = \frac{(6-12)^2 + (9-12)^2 + (12-12)^2 + (15-12)^2 + (18-12)^2}{5-1} = \frac{90}{4} = 22.5$$

Using the alternative formula with $(n-1)$ as divisor

$$s^2 = \frac{(6^2 + 9^2 + \dots + 18^2) - 5(12^2)}{5-1} = \frac{90}{4} = 22.5$$

And the standard deviation is

$$s = \sqrt{22.5} = 4.74$$

For a frequency distribution with k classes, the formula for variance is modified as follows:

$$s^2 = \frac{\sum f(x - \bar{x})^2}{n-1} = \frac{\sum fx^2 - n\bar{x}^2}{n-1}$$

For the frequency distribution in Table 7.3 with x as mid-point, compute the variance as follows:

Age group	Frequency f	Mid-values x	Product fx	fx^2
10-19	29	15	435	6525
20-29	87	25	2175	54375
30-39	25	35	875	30625
40-49	9	45	405	18225
50-59	3	55	165	9075
60-69	1	65	65	4225
Total	154	-	4120	123050

$$\bar{x} = \frac{\sum fx}{n} = \frac{4120}{154} = 26.75$$

$$s^2 = \frac{\sum fx^2 - n\bar{x}^2}{n-1} = \frac{123050 - 154(26.75^2)}{153} = 84.01$$

Hence the standard deviation is

$$s = \sqrt{84.01} = 9.16$$

7.4.2 Standard Error

The standard deviation of a sample mean is often called standard error of the mean or simply standard error ($s_{\bar{x}}$). In other words, the term 'standard error' applies to mean, unless otherwise specified. The standard error is defined as

$$s_{\bar{x}} = \sqrt{\frac{\text{Sample variance}}{\text{Sample size}}} = \sqrt{\frac{s^2}{n}} = \frac{s}{\sqrt{n}}$$

Thus, the standard error is

$$s_x = \frac{9.16}{\sqrt{154}} = 0.74$$

The concept of standard error is best understood with reference to a sampling distribution, an analogous counterpart of a frequency distribution. Just as the standard deviation applies to a frequency distribution, a standard error is applied to a sampling distribution.

Uses of standard deviation

A thorough understanding of the use of standard deviation is difficult for us at this stage, unless we acquire some knowledge on some theoretical distributions in statistics. Nevertheless, we shall try to introduce the idea of its use through a few simple examples. The standard deviation of a population (σ) is a measure of the dispersion in the population, while the standard deviation of sample observations (s) is a measure of the dispersion in the distribution constructed from the sample. In both the cases, the standard deviation represents the average variability in a distribution. The greater this variability around the mean of a distribution, the larger the standard deviation. Thus $s=4.5$ indicates greater variability than $s=2.5$.

To conceptualize the use of standard deviation, let us first use the normal distribution to illustrate the use of population standard deviation. The normal distribution is completely defined by its mean (μ) and standard deviation (σ). An important characteristic of normal distribution (more precisely to say of a normally distributed variable) is that 95.45 percent of the observations or measurements have values which are approximately within $\pm 2\sigma$ (to be more precise, 1.96σ) of the mean. Not only it is true for normal distribution, for most distributions that we deal with, the bulk of the distribution, lies within this interval. Within the range of $\mu \pm 1\sigma$, 68.27 percent measurements and within $\mu \pm 3\sigma$, 99.73 percent measurements lie.

Let us clarify the concept by an example. If the mean height of a sample of 120 women is 158 cm and the standard deviation of these heights is 3 cm, it implies that 114 (95%) women are between $158 \pm 2s = 158 \pm 2(3) = 158 \pm 6$ cm, i.e. between 152 cm and 164 cm. In other words, 3 women (2.5%) are shorter than 152 cm and 3 women (2.5%) are taller than 164 cm. This conclusion is valid as long as it does not deviate too far from the normal model.

Another important use of the standard deviation is in measuring the difference of a given observation from the mean. If an observation has a value x , the difference of this value from the population mean μ , i.e. $x - \mu$

can be expressed as a given number of standard deviation (σ) denoted by z . The z value, which is known as the **standardized normal variate**, is given by

$$z = \frac{x - \mu}{\sigma}$$

In other words, the difference between the mean and an observed value is converted into a difference in terms of the standard deviation. This conversion has wide applications in statistical inference.

Very often in statistical studies, we are interested in specifying the percentages or proportions of items in data set that lie within some specified interval when only the mean and standard deviation of the data set are known. The Russian mathematician Tchebysheff discovered that the fraction or proportion of the data set lying between any two values symmetric about the mean is related to the standard deviation. This rule applies to all distributions, skewed or otherwise. The rule is expressed by the following inequality:

$$P\{|x - \bar{x}| \leq ks\} \geq 1 - \frac{1}{k^2}$$

For $k=2$ the theorem states that at least $1 - 1/2^2 = 75\%$ of the observations must lie within two standard deviations from the mean. That is, 75% or more of the observations of any distribution lie in the interval $\bar{x} \pm 2s$.

The use of standard deviation is manifold. It employs the mathematically acceptable procedure of clearing the signs, and because of this reason, it has wider application than the mean deviation. As a result, the standard deviation has become the initial step for obtaining certain other statistical measures, especially in the context of statistical decision making.

7.4.3 Range

The **range** is the difference between the largest and the smallest score in the distribution. Unlike the standard deviation, it is computed from only the minimum and the maximum scores, and thus it is a very rough measure of variation. Thus for the data set 3, 15, 46, 64, 126, 623, the range is $R=623-3=620$. For the data in Table 7.1, $R=65-12=53$. With the range as a basis of comparison, it is possible to get an idea of the homogeneity (small standard deviation) or heterogeneity (large standard deviation) of the distribution. It is empirically true that for homogeneous distribution, the ratio of the range to the standard deviation should be between 2 and 6. A number above 6 would indicate a high degree of heterogeneity. The range provides useful but limited information for all data.

7.4.4 Inter-quartile Range

The inter-quartile range (IQR) is the difference between the first quartile (Q_1) and third quartile (Q_3) of a distribution. We have already discussed what quartile is.

The minimum value of the distribution when it is divided into 100 equal parts is the 1st percentile (P_1); the maximum the 99th percentile (P_{99}). The first quartile (Q_1) is the 25th percentile, while the third quartile Q_3 is the 75th percentile. The inter-quartile range is thus $IQR = Q_3 - Q_1$. The median or Q_2 is the 2nd quartile. In box plots, Q_1 and Q_3 are called lower and upper hinges respectively. Bowley, a renowned statistician, uses quartile values to assess the skewness of a distribution as follows:

$$\text{Skewness} = (Q_3 - Q_2) - (Q_2 - Q_1)$$

Based on this relationship, we use the following criteria to assess the skewness of a distribution:

- If $Q_3 - Q_2 = Q_2 - Q_1$, the distribution is symmetrical
- If $Q_3 - Q_2 > Q_2 - Q_1$, the distribution is positively skewed
- If $Q_3 - Q_2 < Q_2 - Q_1$, the distribution is negatively skewed

7.4.5 Quartile Deviation

The quartile deviation or semi-interquartile range, is expressed as

$$Q = \frac{Q_3 - Q_1}{2}$$

For a symmetric distribution, Q_3 and Q_1 are equidistant from the median. In such case, the "median $\pm Q$ " encompasses 50% of the observations. Eight Qs cover approximately the range. Q's relationship with the standard deviation (s) is constant ($Q = 0.6745s$) when scores are normally distributed.

7.4.6 Coefficient of Variation

Coefficient of variation abbreviated CV, is a relative measure of dispersion that represents the spread of the distribution relative to the mean of the same distribution. When the means of two or more data sets vary considerably, we do not get an accurate view of the relative variability in the sets just by comparing the standard deviations. Coefficient of variation tends to overcome this difficulty. This is defined as follows:

$$\text{Coefficient of variation} = \frac{\text{Standard deviation}}{\text{Mean}}$$

When expressed in percentage, a value of 40% for the coefficient of variation, for instance, implies that the standard deviation of the sample value is 40% of the mean of the distribution.

The CV is a very useful measure when:

- The observations are gathered in different units for the same variable (such as price of a commodity in dollar and pound)
- The observations are in the same units, but the means are significantly different such as the incomes of rich and poor people).

Consider the following data set to understand the importance of coefficient of variation:

Example 7.15: Suppose *A* and *B* are two companies who have been listed on the Dhaka Stock Exchange. A potential investor is considering purchasing shares in one of the companies. Suppose that each share of stock in company *A* has averaged 5000 taka over the past few months with a standard deviation of 700 taka, while in the same time period, the price per share for company *B* stock averaged 1250 taka with a standard deviation of 250 taka. How can the investor determine which stock is more variable?

Solution: We examine the averages and variability in the share prices of the two companies in the accompanying table

	Company A	Company B
Mean (Tk.)	5000	1250
SD (Tk.)	700	250
CV	14%	20%

The share price of company *A* seems to be more volatile than that of company *B* shares as it is evident from the actual standard deviations. However, because the average prices per share for the two stocks are so different, it would be more appropriate for the potential investor to consider the variability in price relative to the average price. For company *A*, the CV is 14%, while it is 20% for company *B*. Thus relative to the mean, the price of stock *B* is more variable than the price of stock *A*.

Example 7.16: The blood pressures of a group of patients were measured at two levels: systolic and diastolic as shown below.

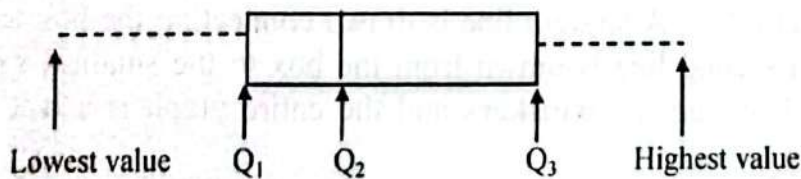
	Systolic	Diastolic
Mean	130 mm Hg	60 mm Hg
SD	15 mm Hg	8 mm Hg
CV	11.5%	13.3%

As the data show, systolic pressure is more variable than the diastolic pressure as shown by the standard deviations. However, in relative terms, as measured by the coefficient of variation, the diastolic pressure has the greater variability.

Both the examples demonstrate that the relative variability is of more concern than absolute variation—hence the importance of CV.

7.4.7 Box and Whisker Plot

The **box and whisker** plot or simply **box-plot** is another technique used frequently in exploratory data analysis. A box plot reduces the details of the stem and leaf plot and provides a different visual image of the distribution's location, spread, shape, tail, length and outliers. Box plot uses the largest and smallest observations, the lower and the upper quartile values, and the median of the distribution in its construction. These quantities are known as the **five-number summary** of a distribution. The plot also demonstrates the concentration of the values in the tails of the distribution. The following diagram illustrates how a box and whisker plot is drawn.



The box extends from Q_1 to Q_3 representing the inter-quartile range and so encloses the 50% of the values. The **whiskers** are the lines that extend from the box to the highest and lowest values (excluding outliers¹) and thus illustrate the range. A line across the box indicates the median (Q_2). The edges of the box are known as **hinges**, which are approximated by Q_1 and Q_3 values. The points lying beyond 1.5 times the inter-quartile range (i.e. above Q_3 and below Q_1) are known as **outliers**. That is

$$\text{Outlier} > Q_3 + 1.5 (Q_3 - Q_1)$$

Or

$$\text{Outlier} < Q_1 - 1.5 (Q_3 - Q_1)$$

Box and whisker plots also convey to us an impression of the location or centering through a special measure of central tendency known as **trimean** and is defined as

¹ Outliers are the values of a distribution which are unusually different (too high or too low) from other values of the distribution

$$\text{Trimean} = \frac{\text{Lower hinge} + 2(\text{median}) + \text{upper hinge}}{4}$$

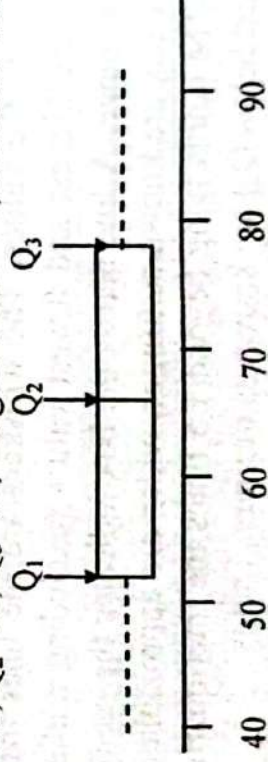
Sometimes it is worthwhile to have some rules about which values deserve to appear individually in a box and whisker plot. Any presentation following these rules gives rise to what is known as **schematic plot**. Specifically, a schematic plot is a representation of a batch, including a hinge box, bared at the median with dashed whiskers extending to the adjacent values terminated by (dashed) bars, and with all outside values identified.

Example 7.17: The following are the number of miles traveled by 51 people by car in a given week. Display the data in a box and whisker plot.

67	52	46	50	78	72	62	93	78	41	47
93	66	56	63	66	81	58	77	78	85	
80	94	70	68	43	42	82	48	44	44	
70	57	58	85	52	72	87	54	48	47	
60	76	74	52	67	86	63	74	53	86	

Solution: The plot is constructed by drawing a box between the lower and upper quartiles, i.e. Q_1 and Q_3 with a solid line drawn across the box to locate the median (Q_2). A straight line is drawn connecting the box to the largest value; a second line is drawn from the box to the smallest value. These straight lines are the **whiskers** and the entire graph is a box and whisker plot.

Here $Q_1=52$, $Q_2=66$, $Q_3=78$, Largest value=94, Smallest value=41.



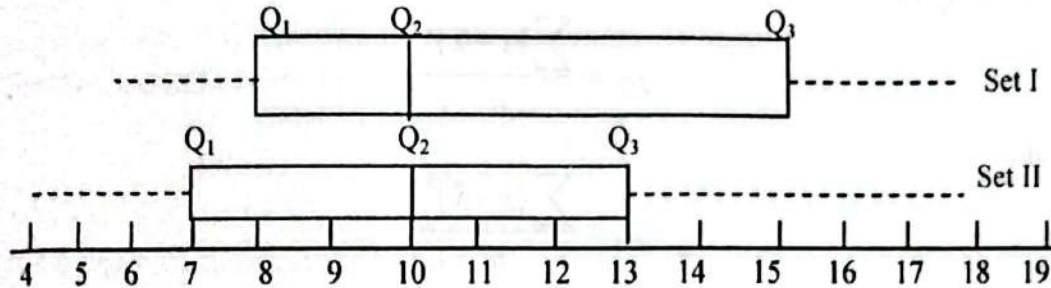
The box plot gives us the immediate impression of an approximately symmetrical distribution with the middle 50% of the values lying between 52 and 78 resulting in an inter-quartile range of 26. The most extreme values are seen to be 41 and 94.

Example 7.18: Given the following information obtained from two sets of data. Draw a box plot to represent these data and comment on the distributions.

Set 1: Median=10, Lower quartile = 8, Upper quartile = 15, Lowest value =6, Highest value = 19.

Set II: Median=10, Lower quartile = 7, Upper quartile = 13, Lowest value = 4, Highest value = 16.

Solution:



The median for both sets is the same. However, the values in the set II are more evenly distributed with a smaller range. There is a bigger spread of values for set I and the distribution for this set is positively skewed.

The illustrations demonstrate that a box plot provides us with the following information:

- ❑ The center of the distribution of scores or values is indicated by the median line in the box plot;
- ❑ A measure of the variability of the values is given by the inter-quartile range;
- ❑ By examining the relative position of the median line, we can guess the symmetry of the middle 50% of the values.
- ❑ Additional information about skewness is obtained from the lengths of the whiskers; the longer one whisker is relative to the other one, the more skewness there is in the tail with the longer whisker.

A general assessment can be made about the presence of outliers by examining the number of values classified as mild outliers and the number classified as extreme outliers.

7.5 Measures of Shape

The measures of shape, viz. skewness and kurtosis, describe departures from the symmetry of a distribution and its relative flatness (or peakedness) respectively. They are related to statistics, known as moments, which use deviation scores ($x - \bar{x}$). The variance, for example, is a second power moment. The measure of skewness uses second and third powered deviations for its computations as follows:

$$s_k = \frac{\mu_3}{\mu_2^{3/2}}$$

Here μ_2 and μ_3 are called second and third moments of a distribution. They are computed as follows:

$$\mu_2 = \frac{\sum (x - \bar{x})^2}{n}$$

and

$$\mu_3 = \frac{\sum (x - \bar{x})^3}{n}$$

Clearly, μ_2 is the variance of a set of observations.

7.5.1 Skewness

Skewness is a measure of a distribution's deviation from symmetry. In a symmetrical distribution, the mean, median and mode are in the same location. A distribution that has cases elongated toward one tail or the other is said to be skewed. When the tail stretches to the left, to smaller values, it is negatively skewed, in which case, $s_k < 0$. Observations stretching toward the right, to larger values, make the distribution positively skewed, in which case $s_k > 0$. For symmetric distribution, $s_k = 0$. The value of μ_3 directly depicts whether the distribution is skewed or not. A distribution will be positively skewed when $\mu_3 > 0$ and negatively skewed $\mu_3 < 0$. When $\mu_3 = 0$, the distribution has no skewness. The following diagrams display different forms of skewness in frequency distributions:

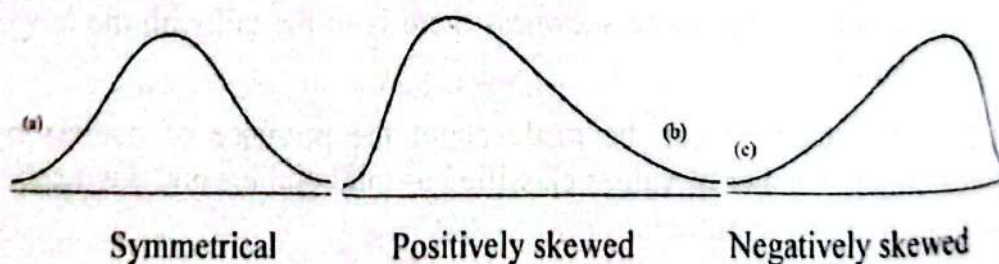


Figure 7.7: Figures showing different forms of skewness

Computation of moments is somewhat cumbersome. One can however check the skewness of a distribution simply examining the mean and the mode if they exist. Thus in general when $\text{mean} > \text{median}$ or $\text{mean} > \text{mode}$ the distribution is positively skewed. Similarly, when $\text{mean} < \text{median}$ or $\text{mean} < \text{mode}$, the distribution is negatively skewed. The distribution presented in Table 7.3 has $\text{mean} = 26.75$, $\text{median} = 25.50$ and $\text{mode} = 24.80$. It is therefore positively skewed. The histogram of this distribution shows

elongated to the right, confirming that the distribution is positively skewed. It is however advisable that the shape of a frequency distribution should be assessed by examining a plot of the frequency distribution (e.g. stem and leaf plot, histogram) not by comparing mean and median.

A bit more refined measure of skewness is due to Pearson, called Pearson's coefficient of skewness and is defined as follows

$$s_k = \frac{\bar{x} - M_0}{s}$$

If the value of the mode is not readily obtained, the difference between the median and the mode can be approximated by the following relationship:

$$\bar{x} - M_0 = 3(\bar{x} - M_e)$$

and in that case, the Pearson's coefficient is expressed as

$$s_k = \frac{3(\bar{x} - M_e)}{s}$$

We emphasize that the above relationships are valid only when dealing with moderately skewed distribution.

7.5.2 Kurtosis

Kurtosis is a measure of a distribution's flatness. Distributions where scores cluster heavily or pile up in the center (along with more observations than normal in the extreme tails) are **leptokurtic**. Flat distributions with scores more evenly distributed and tails flatter than normal distribution are called **platykurtic**. Intermediate or **mesokurtic** distributions are neither too peaked nor too flat. Kurtosis of a distribution is computed as follows:

$$k_u = \frac{\mu_4}{\mu_2^2}$$

where μ_4 is called the 4th powered moment and is computed as follows:

$$\mu_4 = \frac{\sum (x - \bar{x})^4}{n}$$

The value of k_u for a normal or mesokurtic distribution is 3. That is if

$k_u - 3 = 0$, the distribution is mesokurtic

$k_u - 3 > 0$, the distribution is leptokurtic

$k_u - 3 < 0$, the distribution is platykurtic

The distribution, presented in Tale 7.3, has the following values of moments:

$$\mu_2 = 83.29, \mu_3 = 940.46, \mu_4 = 37136.65$$

so that

$$s_k = \frac{\mu_3}{\sqrt{\mu_2^3}} = \frac{940.46}{\sqrt{83.29^3}} = 1.24$$

and

$$k_u = \frac{\mu_4}{\mu_2^2} = \frac{37136.65}{83.29^2} = 5.53$$

Hence the given distribution is slightly skewed to the right since s_k is positive and is leptokurtic since $k_u - 3 > 0$. Examine the accompanying graphs:

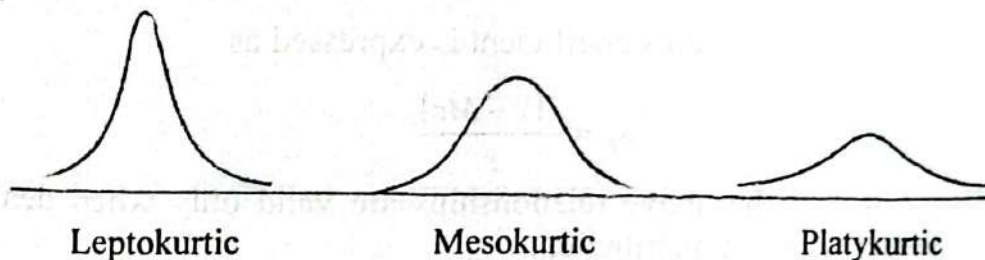


Figure 7.8: Graphs displaying the various forms of kurtosis

7.6 Normal Distribution

Normal distribution is a probability distribution derived by Gauss as a distribution of the error of measurements. This is for this reason is also known as the **normal law of error**. Subsequently, astronomers, physicists, and somewhat later field researchers in wide variety of fields found that their histograms exhibited the common feature of first rising gradually in height to a maximum and then decreasing in a symmetric manner. Although the normal curve is not unique in exhibiting this feature, it has been found to provide a reasonable approximation in great many situations. It is for this reason, normal distribution plays a central role in statistics, and inferential procedures derived from it have wide applicability and form the backbone of current methods of statistical analysis. The curve of the normal distribution is bell-shaped with most of the values clustered near the mean and a few values near the tails. The normal distribution is symmetrical around the mean. A typical normal curve is as shown below:

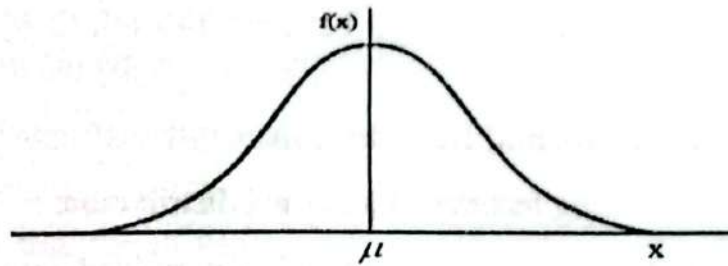


Figure 7.9: A graphical representation of a normal distribution

The mathematical form of a normal distribution is as follows:

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(x-\mu)^2}, \quad -\infty < x < \infty (*)$$

where μ and σ^2 are the mean and variance of the normal distribution.

The normal distribution has some appealing properties. The most important one is that many variables have a normal distribution. The curve of this distribution is bell-shaped with most of the values near the mean and a few values out near the tails. The mean, the median and the mode of a normal distribution have the same value.

7.6.1 Empirical Rule for Normal Distribution

The empirical rule suggests that regardless of the value of the mean or the standard deviation,

- 68.3% of all observations lie within one standard deviation of the mean.
- 95.5% of all observations lie within two standard deviations of the mean.
- 99.7% of all observations lie within three standard deviations of the mean.

Example 7.19: Suppose that a group of 200 women have a mean height of 158 cm with a standard deviation of 3 cm. Assume that the height of the women has an approximately normal distribution. Using the above empirical rules, we can make the following assertions:

- 68.3% women have height between 155 (=158-3) cm and 161 (=158+3) cm.
- 95.5% women have height between 152 (=158-6) cm and 164 (=158+6) cm.

- 99.7% women have height between 149 (=158-9) cm and 167 (=158+9) cm.

7.6.2 Features of a Normal Distribution

Note down the following features of a normal distribution:

- A normal distribution is completely determined once its mean and variance are known. This means that once the mean and variance of a normal distribution are given, then $f(x)$ can easily be calculated for different values of x .
- Two or more normal distributions might have the same variance but different means
- Two or more normal distributions might have different means and different variances.
- In all cases stated above the area under the normal curve will remain 1.
- The range of the normal variate will always vary from $-\infty$ to $+\infty$ irrespective of the choice of the mean and the variance.

7.6.3 Standard Normal Distribution

From the discussion above, we might infer that there can exist an infinite number of possible normal distributions, each with its own mean and variance. Since we obviously cannot examine such large number of possibilities, we need to convert all these normal distributions into one standard form. This form of the distribution is known as the **standard normal distribution**. This conversion is effected through a simple transformation of the x values to z values defined as

$$Z = \frac{X - \mu}{\sigma}$$

The transformed Z defined above is known as the **normal deviate**. The Z value is a standardized value of X and sometimes is known as the standard normal variate and its distribution is standard normal distribution referred to above. Its mean and variance are respectively 0 and 1. The density function of the standard normal distribution is

$$f(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2}, \quad -\infty < z < \infty$$

It is important to remember that all normal distributions can be converted into this standard form. It inherits the same properties the normal

distribution does. Note that the horizontal axis is now measured in units of σ and the distribution is now centered at $\mu = 0$.

16.3.1 Uses of standard normal distribution

The solution of a normal integral such the one in (*) is almost impossible and a hopeless task for all values of the mean and variance. The standard normal distribution is a neat solution to this problem, because it has a unique mean (0) and unique variance (1). What we need to do is to standardize the X values following the rule

$$Z = \frac{X - \mu}{\sigma}$$

To facilitate the computation of normal probabilities, a statistical table has been prepared based on the standard normal distribution. We illustrate the use of this table by a few examples.

Example 7.20: The Philips Bangladesh manufactures electric bulbs that have a length of life that is normally distributed with mean equal to 800 hours and a standard deviation of 40 hours. Find the probability that a bulb selected at random will burn between 810 and 830 hours.

Solution: Let X be the normal variate denoting the length of life of the bulbs. We want to compute the probability $P(810 < X < 830)$. We now standardize two values of X to two Z -values as follows:

$$z_1 = \frac{810 - 800}{40} = 0.25 \text{ and } z_2 = \frac{830 - 800}{40} = 0.75 .$$

This transformation implies that

$$P(810 < X < 830) = P(0.25 < Z < 0.75).$$

Our problem now is to find $P(0.25 < Z < 0.75)$ instead of $P(810 < X < 830)$. We will make use of the attached table (we call it z -table) for the computation of this probability. The z -table is computed from the integral

$$\phi(z) = \int_{-\infty}^z f(z) dz$$

Thus

$$P(0.25 < X < 0.75) = \int_{-\infty}^{0.75} f(z) dz - \int_{-\infty}^{0.25} f(z) dz = 0.7734 - 0.5987 = 0.1747.$$

Hence there is 17.47% chance that the bulb will burn between 810 and 830 hours.

Example 7.21: The Bridgestone tire company from actual road test with tires has estimated the mean tire mileage at 36,000 miles and the standard deviation at 5,000 miles. In addition, the data collected indicate that a normal distribution is a reasonable assumption for the duration of tire life. What percentage of the tires can be expected to last more than 40,000 miles? Less than 20,000 miles?

Solution: We first require computing the probability that X (the tire lasts) exceeds 40,000, that is $P(X > 40,000)$. The corresponding Z value is

$$z = \frac{40,000 - 36,000}{5,000} = 0.70.$$

Hence

$$P(Z > 0.70) = 1 - P(Z \leq 0.70) = 1 - 0.7580 = 0.2420 = 24.2\%$$

Hence a little over 24% of the tires are expected to last more than 40,000 miles.

For the second part of the problem, $z = -3.2$. Hence

$$P(X < 20,000) = P(Z < -3.2) = 1 - P(Z < 3.2) = 1 - 0.9993 = 0.0007,$$

a negligible probability of lasting it less than 20,000 miles.

Example 7.22: The distribution of BMI (Body mass index) in female population aged 20 years and over is assumed to be normally distributed with a mean BMI of 20.4 and a standard deviation of 3.9. What percentage of the women of this population is expected to have a BMI (a) between 18 and 22? (b) above 25?

Solution: For (a),

$$z_1 = \frac{18 - 20.4}{3.9} = -0.62, \quad z_2 = \frac{22 - 20.4}{3.9} = 0.41$$

Hence

$$P(-.62 < Z < .41) = P(Z < .41) - [1 - P(Z < .61)] = .6591 - (1 - .7291) = 0.39$$

For (b)

$$z = \frac{25 - 20.4}{3.9} = 1.18$$

Hence

$$P(Z > 1.18) = 1 - P(Z < 1.18) = 1 - .8810 = 0.12$$

7.7 Confidence Interval

To determine to what extent a particular sample value (say sample mean) deviates from the population value (population mean) a range or an interval around the sample mean can be worked out which will most probably contain the population value (parameter). This range or interval is called the **confidence interval**.

Definition 7.5: A confidence interval is the interval or range of values, which is most likely to encompass the true population value.

The lower and the upper limits of the interval are called **confidence limits**. A 95.5% confidence interval of 152 to 164 cm (for example) for the mean height of a population of women means that we are 95.5% certain that the true population mean lies between 153 and 164. We write this statement as follows:

$$P(152 \leq \mu \leq 164) = 0.955$$

Here 152 is the lower confidence limit and 164 is the upper confidence limit. The calculation of the confidence interval takes into account the standard error (SE) of the estimate (here the mean). The SE gives an estimate of the degree to which the sample mean deviates from the population mean. The value of 95.5% is called the **confidence coefficient**.

Given the standard error of the mean $\sigma_{\bar{x}}$, the 95.5% confidence interval is calculated as follows:

$$\bar{x} \pm 2\sigma_{\bar{x}}$$

The standard error $\sigma_{\bar{x}}$ can be calculated as follows:

$$\sigma_{\bar{x}} = \frac{\sigma_x}{\sqrt{n}}$$

Since the population standard deviation σ_x is unknown in most instances, we simply substitute the sample standard deviation s_x , and obtain the confidence interval. Thus

$$95.5\% \text{ confidence interval for } \mu = \bar{x} \pm 2 \frac{s_x}{\sqrt{n}}$$

We interpret the above interval as follows:

- (a) The probability is 0.955 that the sample mean will be within $\pm 2se$ of the population mean; or

- (b) 95.5% of all the sample means are within ± 2 se from population mean; or
- (c) The population mean is within ± 2 se of 95.5% of all sample means.

The probabilistic statement can be written as follows

$$P\left[\bar{x} - 2\frac{s_x}{\sqrt{n}} < \mu < \bar{x} + 2\frac{s_x}{\sqrt{n}}\right] = 0.955$$

Example 7.23: Eleven children of 3 years of age were weighed and the following weights were obtained:

13, 14, 14, 15, 16, 16, 16, 17, 17, 18, and 20 kg.

Compute the 95.5% confidence interval and interpret your result.

Solution: The mean of the sample values is

$$\bar{x} = \frac{13 + 14 + \dots + 20}{11} = 16 \text{ kg.}$$

The standard deviation is

$$s_x = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}} = \sqrt{\frac{(13-16)^2 + (14-16)^2 + \dots + (20-16)^2}{11-1}} = 2$$

Hence the standard error of the mean is

$$\frac{2}{\sqrt{11}} = 0.6$$

The 95.5% confidence interval for the population mean is

$$\bar{x} \pm 2s_{\bar{x}} = 16 \pm (2 \times 0.6) = 14.8 \text{ to } 17.2$$

This means that we are 95.5% certain that the mean weight of all 3-year old children in the population from which these 11 children have been drawn lies between 14.8 and 17.2.

It is important to note that instead of being 95.5% certain in the estimation, we may be interested to attach a different level of confidence. The most commonly used confidence intervals are 90%, 95% and 99%.

$$90\% \text{ confidence interval for } \mu = \bar{x} \pm 1.64 \frac{s_x}{\sqrt{n}}$$

$$95\% \text{ confidence interval for } \mu = \bar{x} \pm 1.96 \frac{s_x}{\sqrt{n}}$$

$$99\% \text{ confidence interval for } \mu = \bar{x} \pm 2.58 \frac{s_x}{\sqrt{n}}$$

Example 7.24: A large automotive-parts wholesaler needs an estimate of the mean life it can expect from windshield wiper blades under typical driving conditions. The management has selected a simple random sample of 100 wiper blades. For $n=100$, the following values were then calculated from the sample:

$$\bar{x} = 21 \text{ months, } s_x = 6 \text{ months.}$$

Calculate the 95% confidence interval for the population mean and interpret the result.

Solution: The standard error of the mean is calculated as

$$s_{\bar{x}} = \frac{s_x}{\sqrt{n}} = \frac{6}{\sqrt{100}} = 0.6$$

The lower and upper confidence limits for 95% confidence are

$$\mu = \bar{x} \pm 1.96 \frac{s_x}{\sqrt{n}}.$$

That is the,

$$\text{Upper confidence limit} = \bar{x} + 1.96 \frac{s_x}{\sqrt{n}} = 21 + 1.96(0.6) = 22.18 \text{ months}$$

$$\text{Lower confidence limit} = \bar{x} - 1.96 \frac{s_x}{\sqrt{n}} = 21 - 1.96(0.6) = 19.82 \text{ months.}$$

We can now report to the wholesaler with 95% confidence that we estimate the mean life of the wiper blade in the population to be between 19.82 months and 22.18 months.

For sampling without replacement, the 95% confidence interval is

$$\bar{x} \pm 1.96 \frac{s_x}{\sqrt{n}} \sqrt{\frac{N-n}{N}}$$

for which the population size N must be known to us.

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7.8 Correlation Analysis

We now emphasize on the study of the relationship between two or more variables. In dealing with two-variable case, we explore such questions as these:

- Is there a relationship between wages earned and amount of saving?
- Is there a relationship between the amount a firm spends on advertising and sales revenue?
- Is there a relationship between the examination score and the number of hours studied?
- Is there a relationship between age at marriage and years of schooling?

Note that there are two variables in each of the examples cited above. In example (a), wage and the saving are the variables. We expect that savings primarily depend on the wages earned. We therefore refer to saving as the dependent variable and the wage as the independent variable.

In an attempt to analyze the relationship between two variables, we will first define what a correlation analysis is and examine the meaning and intent of correlation analysis by an example.

Definition 7.6: Correlation analysis is the statistical tool we use to describe the degree to which one variable is related to another.

Suppose the counselor of a private university, wants to determine if there is a relationship between the monthly number of follow-up calls and the number of admissions following an advertisement in newspapers. The counselor selects a sample of 10 months from her records. The information is shown in Table 7.5.

Table 7.5: Follow-up and Admission Data

Months	Number of follow-up calls	Number of admissions
1	28	14
2	66	35
3	38	22
2	70	29
5	22	6
6	27	15
7	28	17
8	47	20
9	14	12
10	68	29

If we examine the data, we find that there exists some relationship between the number of calls and the number of students admitted: with the increase in follow-up calls, the number of admissions tends to increase. This type of relationship can be explained and portrayed more precisely by what is known as the correlation analysis

The purpose of correlation analysis is to find how strong the relationship is between two variables without making any distinction between dependent and independent variables. One such measure of this relationship is the **correlation coefficient** and is usually denoted by r . To permit computation of r , both variables are to be measured on a numerical scale (interval or ratio). The value of r can vary from -1 (implying a perfect negative relationship) to $+1$ (implying a perfect positive relationship between the variables under study. A 0 value of r implies that the variables are not linearly related.

Correlation coefficients reveal the magnitude and direction of relationships. The **magnitude** is the degree to which variables move in unison or opposition. The size of a correlation coefficient of .40 for example, is the same as one of $-.40$. The sign of r says nothing about its size. The coefficient's sign indicates the direction of the relationship. We summarize below a few characteristic features of r :

- For any data set, r lies between -1 and $+1$, i.e. $-1 \leq r \leq 1$.
- When $r = +1$, or -1 , the relationship is perfect, that is all points (x, y) exactly lie on the regression line.
- When both the variables move in the same direction (downward or upward), r is positive.
- When the variables move in the opposite directions, r is negative.
- When one variable moves and the other variable remains static, r is zero.
- When r is zero, we might also suspect that the variables are not linearly related. However, we could also obtain $r = 0$ if some curvilinear (non-linear) relationship exists between x and y .
- A useful interpretation of r is that its square (r^2) measures the proportion of the variability in y accounted for by the linear relationship with variable x .

Age and height, for example, are expected to have positive relationship, while prices of products will have a negative relationship with demand. The absence of a relationship is expressed by a coefficient of approximately zero. **Scatterplots** or **scatter diagrams** are a useful tool for

understanding and examining the relationship between variables. A scatter diagram can provide us with two types of information:

- Usually, we can look for patterns that indicate that the variables are related;
- If the variables are related, we can see what kind of line or estimating equation describes such relationship.

Both the direction and the extent of relationship are conveyed in a scatter diagram. The scatter plot for the follow-up calls–admission data are displayed in Figure 7.6.

Note that the pattern of points results because each pair of data from the table has been recorded as a single point. When we view all these points together, we can visualize the nature of relationship that exists between the two variables. As a result we can draw a straight line through our scatter diagram to represent the relationship. Clearly, the relationship appears to be linear and as such the implied correlation is linear correlation.

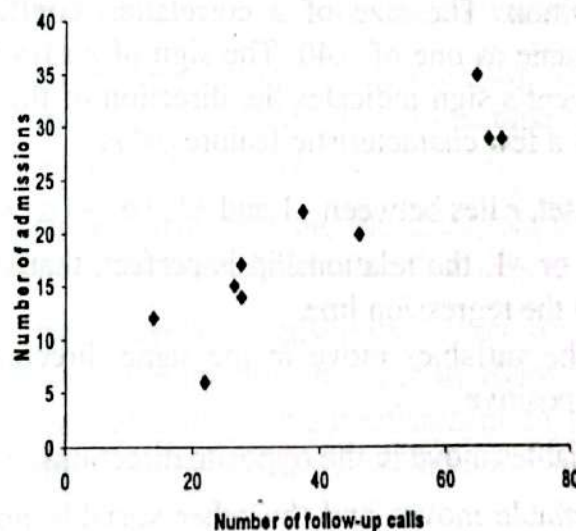


Figure 7.10: Scatter diagram showing the follow-up calls and admission data

When n pairs of values corresponding to two variables x and y are given, one can compute r using the formula below:

$$r = \frac{n \sum xy - (\sum x)(\sum y)}{\sqrt{[n \sum x^2 - (\sum x)^2][n \sum y^2 - (\sum y)^2]}}$$

For the follow up calls-admission data, the value of r can be computed first constructing the following table:

Month	x	y	xy	x^2	y^2
1	28	14	392	784	196
2	66	35	2310	4356	1225
3	38	22	836	1444	484
4	70	29	2030	4900	841
5	22	6	132	484	36
6	27	15	405	729	225
7	28	17	476	784	289
8	47	20	940	2209	400
9	14	12	168	196	144
10	68	29	1972	4624	841
Total	408	199	9661	20510	4681

Using the formula

$$r = \frac{10(9661) - (199)(408)}{\sqrt{[10(4681) - 39601][10(20510) - 166464]}} = \frac{15418}{16689} = .924$$

To interpret this coefficient, we first note that it is positive. So we conclude that there is a direct relationship between the two variables, which means that follow-up calls increase admission. This confirms our reasoning based on the scatter diagram. Second, the value of .924 is close to 1, so we conclude that the association is strong.

When r is squared, we get what we call **coefficient of determination**. Here $r^2 = (.924)^2 = .85$. This value tells us that 85% of the total variations in y (admission) that is explained or accounted for, by the variation in the variable x (calls). The remaining 15% remains unexplained meaning that more variables are needed to explain the variations in y . Hence simple correlation analysis is not enough for the purpose.

Example 7.25: In a nutrition study, a sample of 10 children under 5 years of age was weighed and their daily family incomes recorded. The results were as follows:

Family	Family income in US\$	Weight in kg
1	13	15
2	20	19
3	34	21
4	24	16
5	16	12
6	30	16
7	36	18
8	11	18
9	8	13
10	27	20

Compute r from the given data. Also compute the coefficient of determination (r^2) and interpret your results.

Solution: The computation of r and hence r^2 is facilitated through computing first sum of squares of x , y and sum of product of x and y . We construct the following table to illustrate the computation of r , the coefficient of correlation.

As we see, the computed value of r is 0.60. Since the value is positive, it indicates an upward sloping line but a long way from 1, the maximum value that r can take on. The coefficient of determination is obtained as performing the square root of r :

$$r = \sqrt{.60} = 0.77 = 77\%.$$

Thus the independent variable explains 77% of the total variation in y .

Family	Income x	Weight y	x^2	y^2	xy
1	13	15	169	225	195
2	20	19	400	361	380
3	34	21	1156	441	714
4	24	16	576	256	384
5	16	12	256	144	192
6	30	16	900	256	480
7	36	18	1296	324	648
8	11	18	121	324	198
9	8	13	64	169	104
10	27	20	729	400	540
Total	219	168	5667	2900	3835

$$r = \frac{10(3835) - (219)(168)}{\sqrt{[10(5667) - 47961][10(2900) - 28224]}} = \frac{1558}{2599} = .60$$

7.8.1 Spearman's rho (r_s)

The Spearman's *rho* (r_s) correlation is a popular and widely used ordinal measure. *rho* correlates rank between two ordered variables. To compute r_s , the two variables are each rank ordered, with tie being given the arithmetic mean of the ranks that would have been received. Thus each person or case has two ranks, one on each variable. If the difference between the pairs of ranks for each person is d_i ($i=1, 2, \dots, n$), then for n pairs of observations, the r_s is computed as

$$r_s = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)}$$

The value of r_s varies from +1 when all pairs of ranks are identical; -1 when no pairs are identical. When $r_s = 0$, no relationship exists between the two variables (so that the ranks of one variable are in complete disagreement with the ranks of the other variable). Example 7.26 illustrates the computation of r_s .

The computed correlation coefficient implies that the relationship between the two sets of rankings is rather weak, suggesting lack of agreement between the measures. The computed rank correlation can also be tested for its statistical significance. This has been taken up in a separate chapter.

Example 7.26: The systolic blood pressures of 5 individuals were measured by two physicians and were then ranked. The measurements were as follows:

Individual	Physician 1	Physician 2
1	120	129
2	118	110
3	140	138
4	148	120
5	145	150

Compute the rank correlation coefficient between the measurements of blood pressures by the physicians.

Solution: To compute r_s , we need to rank order the measurements. This is done in the accompanying table.

Individual	Physician 1	Physician 2	Difference (d_i)	d_i^2
1	4	3	1	1
2	5	5	0	0
3	3	2	1	1
4	1	4	-3	9
5	2	1	1	1
Total	-	-	0	12

The required rank correlation coefficient is

$$r_s = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)} = 1 - \frac{6 \times 12}{5(5^2 - 1)} = 0.40$$

It is of importance to note that ranking of the measurements can be done directly without going into the intermediate steps of numerical measurement as above.

7.9 Regression Analysis

We now turn to develop an equation to express the relationship between two variables. In addition, we want to estimate the value of the dependent variable (y) based on a fixed value of the independent variable (x). The technique used to develop the aforesaid equation and making this prediction is called **regression analysis**. The resulting equation is referred to as the **regression equation**. The regression equation is a mathematical equation that defines the relationship between the dependent variable and the independent variable.

If you look at the relationship between the family income and the weight of the children, you will see that children from low income families had lower weight, on average, than children from families with high incomes. This can also be examined by constructing scatter diagram and inspecting simply by eye. The upward trend in weight with increasing family income will be clearly seen in the diagram. However, examining by eye is rather subjective and we would prefer a more objective technique. This is the idea behind regression analysis. We accomplish this analysis using a method known as **least squares method** that gives the best line of prediction of the dependent variable for any pre-assigned value of the independent variable.

Any straight line drawn on a graph paper can be represented by the straight line of the form

$$\hat{y} = a + bx$$

where

$\hat{y}$: Predicted or estimated value of y for a fixed value of x .

a : y -intercept which represents the estimated value of y when $x=0$

b : Slope of the line or average change in $\hat{y}$ for each one unit change in x .

It is also known as the regression coefficient of y on x .

x : Any value of the independent variable that is fixed in advance.

Each point on the line has an x value and a y value, and the equation tells us how these x and y values are interrelated. For different values of a and b we will have different straight lines.

To decide on the line to fit through our scatter, we have to decide what values of a and b to use. In least-squares method, basically, we choose

and b in such a way that the vertical distances of the points from the line are minimized. To be more precise, we choose a and b that minimize the squares of these vertical distances. Hence the name 'least-squares method'. The detail of the method is beyond the scope of this text. We simply illustrate how the constants a and b can be computed from the observed data and interpret them

In the language of regression analysis, the constant b in the regression set up

$$\hat{y} = a + bx$$

is called the **coefficient of regression of y on x** , and a , the regression constant. The coefficient b is computed first using the formula

$$b = \frac{n \sum xy - (\sum x)(\sum y)}{n \sum x^2 - (\sum x)^2}$$

Having estimated b , the constant a is computed from the observed data using the formula

$$a = \frac{\sum y}{n} - b \frac{\sum x}{n} = \bar{y} - b\bar{x}$$

In example 7.25, our estimate of b is

$$b = \frac{10(3835) - (219)(168)}{10(5667) - 47961} = \frac{1558}{8709} = .1789$$

The estimate of a is

$$a = \frac{\sum y}{n} - b \frac{\sum x}{n} = \frac{168}{10} - .1789 \times \frac{219}{10} = 12.88$$

The equation of our fitted or estimated regression line is thus

$$\hat{y} = 12.88 + 0.1789x$$

The regression line calculated above is of immense importance. To realize this, first construct scatter diagram with the observed values of x and y on a graph paper and then estimate y values using each value of x from the fitted regression line. With these estimated values, draw a new line on the same graph paper. This line is your least squares line. Now examine this line. You see that all the estimated values lie on this new line. The differences between the observed and the estimated values are the errors due to estimation. The accompanying table displays the observed and the estimated values of y .

x	y	$\hat{y}$
13	15	15.2
20	19	16.4
34	21	19.0
24	16	17.2
16	12	15.7
30	16	18.2
36	18	19.3
11	18	14.8
8	13	14.3
27	20	17.7

The estimated line is also called **prediction line**. You can use this line to predict y for any intermediate values of x . You can also make extrapolation beyond the given values in some cases. For example, you might be interested to infer how much a child is expected to weigh with family income \$25. To answer this you simply need to put $x=25$ in the above equation and obtain:

$$\hat{y} = 12.88 + .1789(25) = 17.35$$

For $x=40$ (when we are extrapolating),

$$\hat{y} = 12.88 + .1789(40) = 20$$

It is however dangerous to extrapolate the regression line outside the range of the data. For example, extrapolating the line to an income of 250 would yield an estimated mean weight of

$$\hat{y} = 12.88 + 0.1789(250) = 57.6 \text{ kg,}$$

which is of course absurd for a child under 5. With no family incomes, the estimated weight remains at 12.88 kg, as implied by the value of a .

How do we interpret the regression line? The regression line, as we see above, estimates the **average** value of y for a given value of x . In the above example, we see that children whose families have an income of \$25 would on average weigh 17.35 kg, though some would weigh more and some less than this.

The regression coefficient b tells us the increase in the average value of y corresponding to a unit increase in x . In our example, mean weight increased by 0.1789 kg (or about 179 gram) for each increase of \$1 in family income.

Example 7.27: Determine the regression equation for the follow-up calls data and interpret the constants a and b .

Solution: You can check that for the given data

$$b = \frac{10(9661) - (408)(199)}{10(20510) - (408)^2} = \frac{15418}{38636} = .40$$

The estimate of a is

$$a = \frac{\sum y}{n} - b \frac{\sum x}{n} = \frac{199}{10} - .40 \times \frac{408}{10} = 3.58$$

so that our estimated regression line is

$$\hat{y} = 3.58 + 0.4x$$

So if a counselor makes 20 calls a month, the expected number of admissions is about 12:

$$\hat{y} = 3.58 + 0.4(20) = 11.58$$

The b value tells us that each additional call made during the month, is expected to increase 0.4 admissions on average, or in other words, 4 admissions for each 10 calls. The value of 3.58 indicates that if no calls were made, approximately 4 admissions may be expected.

7.9.1 Standard Error of Estimate

The standard error of the estimate is a measure that indicates how precise the prediction of y is based on x or, conversely, how inaccurate the prediction might be. The standard error of the estimate, denoted by s_e , is the same concept as the standard deviation we discussed earlier in this chapter. The standard deviation measures the dispersion about an average, while the standard error of the estimate measures the dispersion about an average line, the regression line.

The standard error of the estimate is computed as follows:

$$s_e = \sqrt{\frac{\sum (y - \hat{y})^2}{n - 2}}$$

We determine below the standard error of the estimate for the nutrition survey data in Example 7.25

x	y	$\hat{y}$	$(y - \hat{y})^2$	y^2	xy	$(y - \bar{y})^2$
13	15	15.2	.04	225	195	3.24
20	19	16.4	6.76	361	380	4.84
34	21	19.0	4.00	441	714	17.64
24	16	17.2	1.44	256	384	0.64
16	12	15.7	13.69	144	192	23.04
30	16	18.2	4.84	256	480	0.64
36	18	19.3	1.69	324	648	1.44
11	18	14.8	10.24	324	198	1.44
8	13	14.3	1.69	169	104	14.44
27	20	17.7	5.29	400	540	10.24
Total	168	—	49.68	2900	3835	76.96

The standard error of the estimate is thus

$$s_e = \sqrt{\frac{\sum (y - \hat{y})^2}{n - 2}} = \sqrt{\frac{49.68}{8}} = 2.50$$

In most cases, the standard error is computed using an alternative formula

$$s_e = \sqrt{\frac{\sum y^2 - a \sum y - b \sum xy}{n - 2}}$$

To make use of this formula, we need an additional column containing the product of x and y . This column is shown in the above table. With the estimated values $a=3.58$ and $b=.40$, the alternative formula yields

$$s_e = \sqrt{\frac{2900 - 12.88(168) - .1789(3835)}{8}} = 2.50$$

as ought to be.

Relationship between r^2 and standard error of estimate

To demonstrate the relationship we partition the total variation in the regression set up as follows:

$$\text{Total variation} = \text{Regression} + \text{Error variation}$$

In terms of sum of squares

$$\sum (y - \bar{y})^2 = \sum (\hat{y} - \bar{y})^2 + \sum (y - \hat{y})^2$$

The terms are called total sum of squares (SST), regression sum of squares (SSR) and error sum of squares (SSE) respectively. That is

$$SST = SSR + SSE$$

The coefficient of determination, r^2 can now be obtained using the following relationship:

$$r^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST} = 1 - \frac{\sum (y - \hat{y})^2}{\sum (y - \bar{y})^2}$$

To compute r^2 from the nutrition data, we incorporate an additional column for SST in the above table. Thus

$$r^2 = 1 - \frac{\sum (y - \hat{y})^2}{\sum (y - \bar{y})^2} = 1 - \frac{49.68}{76.96} = 0.354$$

Consequently, the correlation coefficient is $r = \sqrt{.354} = .60$, which agrees with our previous computation. Note that the standard error of estimate in terms of SSE can be computed using the formula

$$s_e = \sqrt{\frac{SSE}{n-2}}$$

7.10 Multiple Regression Analysis

It is generally the case in all scientific investigations that there is no single cause of a given phenomenon. The inherent complexity of most real world problems necessitates the consideration of more than one influencing variable in order to predict the dependent variable. As an example, the demand for a commodity is likely to be dependent not only on its own price, but also on the prices of other competing or complementary goods, income of consumers and the like. The yield of wheat may depend on such factors as fertilizer, rainfall, soil fertility, irrigation etc. Therefore we need to extend our simple 2-variable regression to cover regression involving more than two variables. In addition, there are many important situations, where the underlying relationship between two variables cannot be explained adequately with a straight-line relationship. In such situations, our two-variable regression analysis, as discussed earlier, appears inadequate. This, in turn, leads to the consideration of **multiple regression analysis**, in which the dependent variable depends on two or more explanatory variables.

Let us try further to understand the necessity of considering multiple regression with an example. If expenditure (y) is correlated with a single independent variable - income (x_1), and a correlation of $r = 0.5$ is found, this will mean that $r^2 = 0.25$ and that x_1 explains only one-fourth of the total variance in y . In order to explain the remaining 75 percent of the variance, we might add additional variables to the analysis. We can add such variable as family size to the analysis and label this variable as x_2 . The resulting observed regression function is of the form:

$$y = a + b_1x_1 + b_2x_2 \quad (a)$$

The equation above is the simplest form of a multiple regression equation and the analysis involved here is known as the **multiple regression analysis**. The b values in the regression set-up (a) are referred to as **partial regression coefficients**. They are called partial because each of the coefficients (b_1, b_2) indicates only the effects of its respective independent variable on the dependent variable, with the effects of all other independent variables in the equation statistically controlled for. Thus the equation is an additive one, with each regression coefficient indicating an independent and unique effect not shared by other variables in the equation. The regression coefficients referred to above give the change in the dependent variable (in whatever unit the dependent variable is measured) that we would estimate for a one-unit change in the specified predictor (in whatever unit the predictor is measured). The assumptions in the multiple regression are similar to those of simple regression discussed earlier. Note that this model is just an extension to two independent variables of the one that we initially developed earlier on simple linear regression.

The quantities a, b_1 , and b_2 , need to be computed from available sample data. For the estimation purpose, we can follow principles of least-squares as described in earlier chapter. We can interpret the estimated coefficient a as the y intercept of the estimated regression function. In addition, b_1 , and b_2 can be interpreted as slopes of the independent variables. Needless to say, the equation (a) can be extended for any number of independent variables as follows:

$$y = a + b_1x_1 + b_2x_2 + \dots + b_kx_k \quad (b)$$

To illustrate the use of multiple regression, consider the example below.

Example 7.28: For 10 families, the following data were available on their income (in '000 taka), expenditure (in '000 taka) and family size. We will use these data to compute the multiple regression equation and the standard error of estimate and the coefficient of determination.

Family	Expenditure (y)	Income (x ₁)	Family size (x ₂)
1	7.0	10	4
2	8.0	12	5
3	9.5	15	5
4	10	18	8
5	11	18	7
6	15	20	9
7	18	20	8
8	18	19	9
9	20	20	9
10	25	29	10

Considering income and family size as independent variables and expenditure as a dependent variable, the following three-variable multiple regression equation can be formulated:

$$\hat{y} = a + b_1x_1 + b_2x_2$$

Using a computer software, we computed the values of a , b_1 and b_2 as $a = -5.463$, $b_1 = .734$, $b_2 = .855$, so that the estimated regression line is

$$\hat{y} = -5.463 + .734x_1 + .855x_2$$

A value of -5.463 of ' a ' indicates that the expenditure is -5463 taka when both income and family size are zero. Of course, it does not make any sense to have zero income and zero family size and consequently negative expenditure. It is important to keep in mind that a regression equation is not effective outside the range of the sample values for the dependent variables.

The b_1 of $.734$ implies that for each increase of 1 taka in income, the family expenditure would increase $.734$ taka, regardless of the family size, that is when family size is held constant. The b_2 value of $.855$ implies that for each increase of one member in the family, the family expenditure would increase $.855$ taka regardless of the family income. For a family with 6 members and an income of 17000 taka, the estimated average family expenditure is Tk. 12145:

$$\hat{y} = -5.463 + .734(17) + .855(6) = 12.145$$

To compute the standard error and the coefficient of determination, we prepare the following table:

Family	(y)	$\hat{y}$	$(y - \bar{y})^2$	$(y - \hat{y})^2$
1	7.0	5.30	51.12	2.90
2	8.0	7.62	37.82	0.14
3	9.5	9.82	21.62	0.10
4	10	14.59	17.22	21.06
5	11	13.73	9.92	7.47
6	15	16.91	0.72	3.66
7	18	16.06	14.82	3.78
8	18	16.18	14.82	3.32
9	20	16.91	34.22	9.54
10	25	24.37	117.72	0.39
Total	141.5	-	320.02	52.36

The multiple correlation of determination is calculated as follows:

$$R^2_{y.x_1x_2} = 1 - \frac{SSE}{SST} = 1 - \frac{\sum (y - \hat{y})^2}{\sum (y - \bar{y})^2} = 1 - \frac{52.36}{320.02} = 0.836$$

explaining 83.6% of the variations in y (expenditure) due to income and family size. The multiple correlation coefficient between the dependent variable y and the independent variables x_1 and x_2 jointly is

$$R_{y.x_1x_2} = \sqrt{.836} = 0.915$$

The standard error of the estimate is obtained using the general formula

$$s_e = \sqrt{\frac{\sum (y - \hat{y})^2}{n - (k + 1)}}$$

where k is the number of independent variables in the regression. In our case, $k=2$, so that

$$s_e = \sqrt{\frac{\sum (y - \hat{y})^2}{n - 3}} = \sqrt{\frac{52.36}{10 - 3}} = 2.73$$

EXERCISES 7

1. What is statistics? What are its functions? Discuss in brief the role of statistics in research.
2. What are the various ways of summarizing data? Illustrate with an example the concept of a frequency distribution. How is it constructed from scores?
3. What are qualitative data? What are the different types of diagrams?

presenting qualitative data? Discuss the methods of constructing bar charts and pie charts indicating the type of data required.

- a. The data in the accompanying table represent the water consumption per household in a residential area. Represent the data by a bar and a pie chart.

Reason	Gallon per day
Bathing	100
Dish washing	15
Drinking and cooking	10
Lawn watering	150
Miscellaneous	20

- s. Define class width, class mark and class frequency. Given the following frequency distribution of daily wages in dollar of 60 workers:

Wage rates (\$)	Frequency
05-10	2
10-15	4
15-20	9
20-25	15
25-30	21
30-35	6
35-40	3
Total	60

Construct a relative frequency distribution, and a cumulative frequency distribution of both less than and more than type. Determine also the class midpoints, and class widths. What percentages of the workers receive 20 or more dollars a day? Less than 30?

- What is a stem and leaf plot? Discuss its superiority of other methods of data presentation. Compare and contrast a histogram and a stem and leaf plot. Represent the data in table 7.1 by a stem and leaf plot.
- What are five number summaries? What are their usefulness? How are they shown in a box and whisker's plot? Discuss the role of box and whisker's plot in describing the shape characteristics of a frequency polygon.
- What is a relative frequency distribution? How is it constructed? The following data represent the number of applications received for opening new accounts in the bank over a period of 160 days. Construct a frequency distribution with suitable class intervals. Construct also cumulative frequency distributions and relative frequency distribution.

5, 10, 7, 11, 5, 8, 8, 10, 5, 10, 5, 4, 0, 0, 9, 5, 1, 1, 5, 5, 4, 5, 5, 2, 2, 5, 4,
 5, 4, 3, 4, 0, 5, 13, 5, 4, 6, 4, 5, 10, 5, 8, 8, 4, 0, 5, 2, 5, 6, 4, 8, 5, 5, 8, 4,
 5, 2, 5, 6, 7, 5, 5, 4, 7, 5, 5, 8, 14, 7, 8, 12, 15, 5, 12, 7, 15, 14, 16, 5, 4,
 0, 0, 9, 5, 1, 1, 5, 5, 4, 5, 5, 2, 2, 5, 4, 5, 4, 3, 4, 0, 5, 13, 5, 4, 6, 4, 5, 10,
 5, 8, 8, 4, 0, 7, 11, 5, 8, 8, 10, 5, 10, 5, 4, 0, 0, 9, 5, 1, 1, 5, 5, 4, 5, 5, 2,
 2, 5, 4, 5, 4, 7, 5, 8, 8, 10, 11, 3, 4, 0, 5, 13, 5, 7, 11, 5, 8, 8, 10, 5, 10,

9. What is a contingency table? Why is it necessary? How are marginal distribution tables constructed from a cross-table? The following data represent the bivariate responses of 40 customers on their sex (Male =M, Female =F) and types of account they maintain in a bank (Savings=S and Current=C).

Sex:	M	M	F	M	M	F	M	F	M	F	F
Account:	C	C	S	C	S	C	S	C	S	C	S
Sex:	F	F	M	M	F	M	F	M	F	F	F
Account:	S	C	C	S	S	S	S	S	S	S	C
Sex:	F	F	M	M	F	M	F	M	F	F	F
Account:	S	C	C	S	S	S	S	S	S	S	C
Sex:	F	F	M	M	F	M	F	M	F	F	F
Account:	S	C	C	S	S	S	S	S	S	S	C

Tally the data into a 2×2 contingency table and form a contingency table based on 40 responses.

10. A schedule bank conducted a customer satisfaction survey on a monthly basis to measure satisfaction with several areas of services offered by a branch office. The results from a sample of 200 customers were as follows:

Area of services	Number satisfied	Number dissatisfied
Waiting time for tellers	123	65
Automatic	73	7
Investment advisement	43	6
Traveler's check service	25	11
Safe deposits	24	5
Account maintenance services	46	4

- (a) Construct tables of row, column and total percentages.
 (b) Construct a multiple bar chart of customer satisfaction by area of service.
11. What is central tendency of data? What are its measures? Define sample mean and the state the steps in calculating the sample mean from a set of observations.
12. What is median? When is it more preferable to mean? For the last 10 days in July 2007, the Dhaka mail train arrived late at its destination by the times in minutes listed below. (A negative sign means that the train was early by the number of minutes indicated),

−3, 6, 4, 10, −4, 124, 2, −1, 4, 1

Compute the mean, median and mode and interpret your results.

13. What are quartiles and percentiles? What are their practical significance? Given the following frequency distribution of 100 workers by their current age. Calculate the mean, median, mode, 5th percentile and 3rd quartile.

Age group	Number of workers
20-24	3
25-29	10
30-34	19
35-39	35
40-44	18
45-49	10
50-54	5
Total	100

14. Discuss weighted mean and pooled mean. Describe situations when these means are appropriate. A family is composed of 5 males and 3 females. The average weight for the males is 67 kg, while it is 49 kg for the females. What would be the average weight of all the 8 members in the family?
15. In a store, the average monthly sales during a year were as follows:

Period	Average monthly sales in lac
January-March	2.5
April-August	2.9
September-December	3.1

Compute the monthly average sales for the whole year (January-December)

16. A taxicab driver noted that during the last 4 weeks, he purchased 20 gallons of gasoline at 40 taka per gallon; 25 gallons at 45 taka per gallon, 40 gallons at 43 taka per gallon and 30 gallons at take 40 per gallon. What was the average he paid for a gallon of gas?
17. What are the various measures of dispersion? Why are they necessary? Illustrate with example. Define variance and standard deviation. State some properties of variance.
18. The number of ballpoint pens sold in a store over a period of seven days were as follows:

12, 14, 20, 10, 23, 22, 18

Calculate the variance and the standard deviation. Also calculate the coefficient of variation. Compute the standard error of the mean. And hence construct a 95% confidence interval for the population mean. .

19. Compute the variance, standard deviation and the coefficient of variation from the data in Exercise 13.
20. What is a boxplot? What are five number summary? Two sweetmeat shops provided the following information on their daily sales

City	Minimum	Q_1	Median	Q_3	maximum
A	18400	24000	28300	30400	36300
B	19600	26500	31200	35700	41800

- (a) Construct a box plot for the sales volume in A.
 - (b) Construct a box plot on the same graph, for B
21. Define skewness and kurtosis of a distribution. Why are they important? Draw the following curves and locate the approximate positions of mean, median and mode:
 - (i) Symmetric curve, (ii) Positively skewed curve, and (iii) Negatively skewed curve.
 22. How are the mean, median and mode empirically related? How do you infer the skewness of a distribution from such relationships? Describe the Pearsonian coefficient of skewness. State the conditions under which this coefficient is applicable for measuring the skewness of a distribution.
 23. Why is normal distribution important? State the properties of a normal distribution. What are the empirical rules of normal distribution with respect to the area?
 24. Define standard normal distribution and state its properties. The lives of Phillips bulbs are normally distributed with a mean equal to 11350 hrs and a standard deviation equal to 18 hrs.
 - (i) What percentage of bulbs will have a life between 1341 and 1365?
 - (ii) What percentage of bulbs will last longer than 1386 hrs?
 - (iii) What percentage of bulbs will last less than 1323 hrs?
 25. What is confidence interval? How is it constructed from sample data? Illustrate with an example. Interpret the following intervals: $\bar{x} \pm 1s$, $\bar{x} \pm 2s$, $\bar{x} \pm 3s$
 26. Distinguish between correlation analysis and regression analysis indicating their functions in statistical analysis. Discuss the role of dependent variable and independent variable in this context.
 27. What is a scatter diagram? What roles does it play in understanding the relationship between variables? Illustrate with examples. Draw scatter diagrams when $r=0$, $r=-1$, $r=+1$, $r=-.90$ and $r=+.20$
 28. Compare and contrast the properties of correlation coefficient and regression coefficient and present formula to compute them from sample data.
 29. What is rank correlation? Under what circumstances would it make sense to compute rank correlation rather than product moment correlation? Give illustrative examples. Also write down the steps involved in the computation of rank correlation from pertinent data.
 30. What is standard error of estimate? What does it measure? Provide all possible ways of computing this measure.
 31. How does simple regression differ from multiple regression? Illustrate with an example. Write down their general forms. How do you compute the multiple correlation coefficient and standard error of estimate for a three-variate multiple regression set-up?